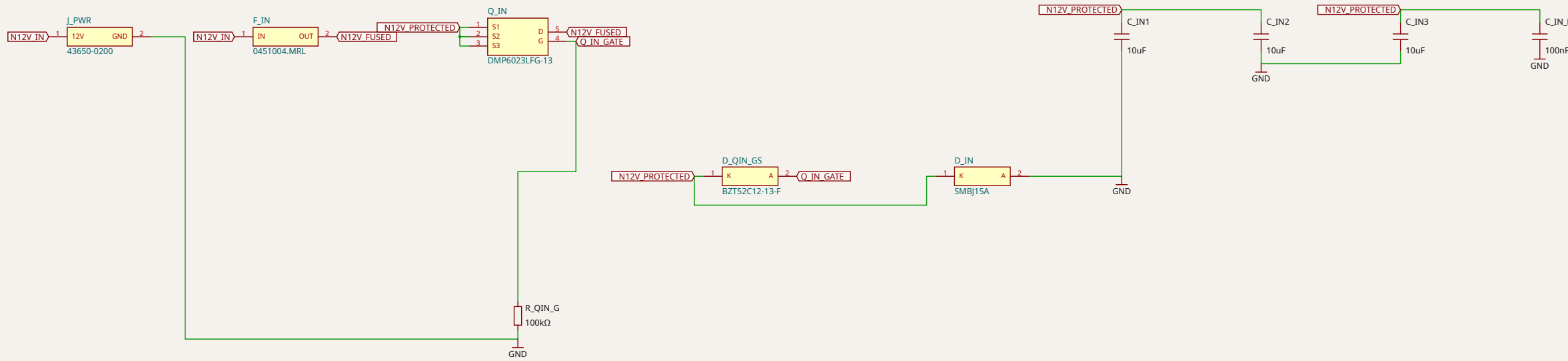


Crow USB Carrier v1

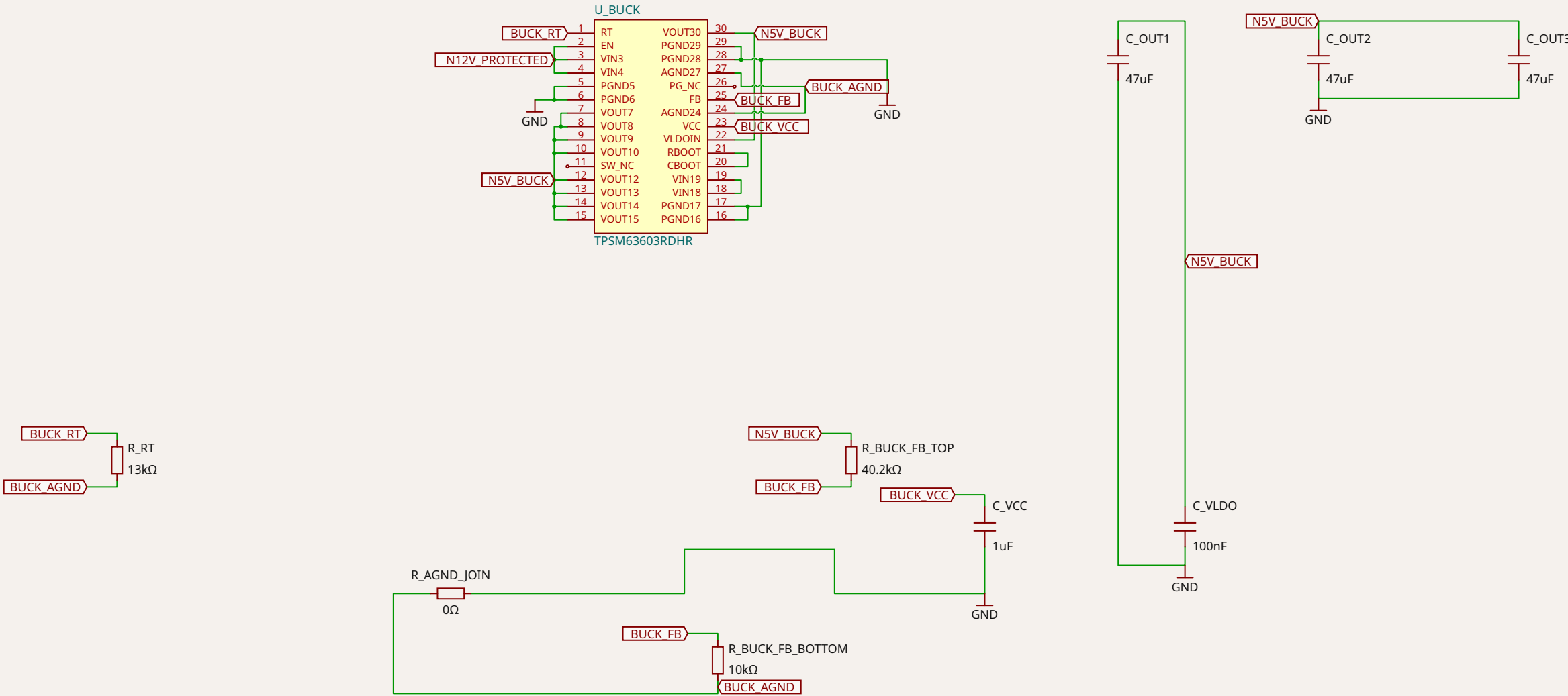
POWER INPUT



Crow USB Carrier v1

POWER BUCK

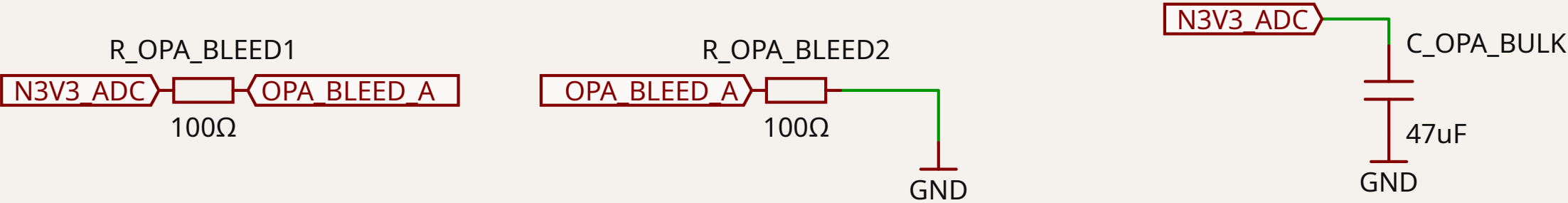
Page 2 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 10 components • landscape page fit



Crow USB Carrier v1

ANALOG POWER AUX

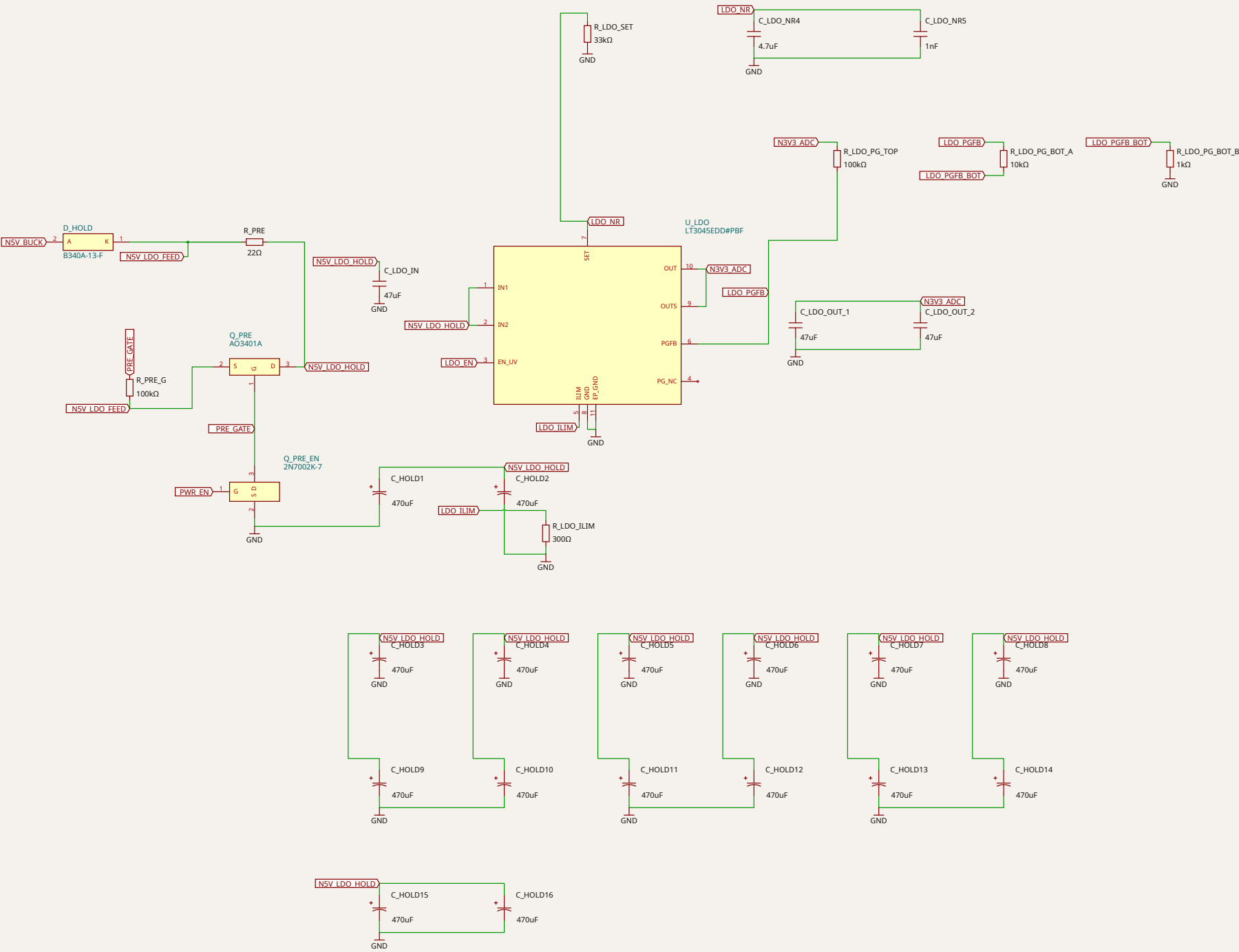
Page 3 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 3 components • landscape page fit



Crow USB Carrier v1

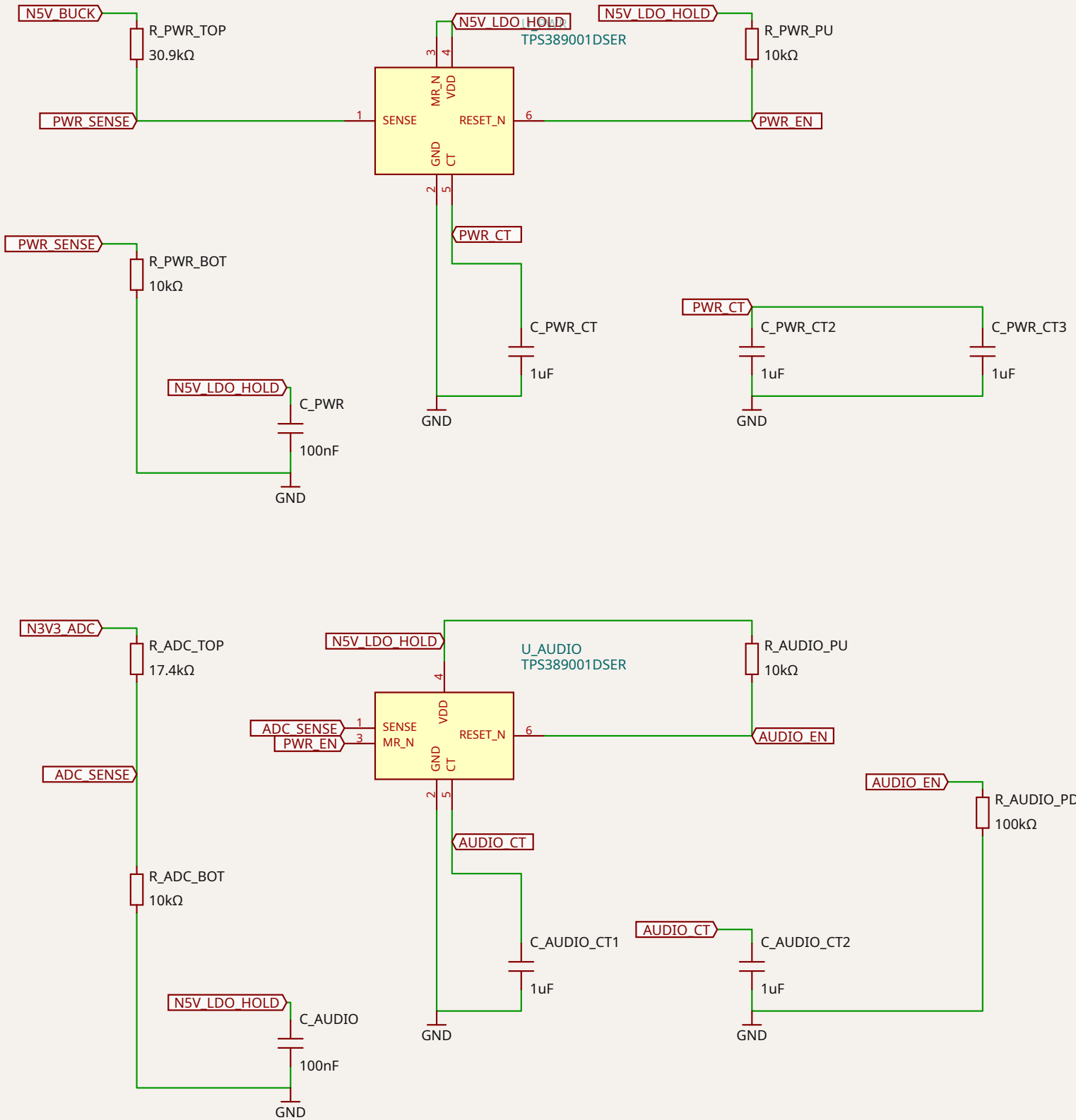
HELD LDO

Page 4 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 32 components • landscape page fit



Crow USB Carrier v1

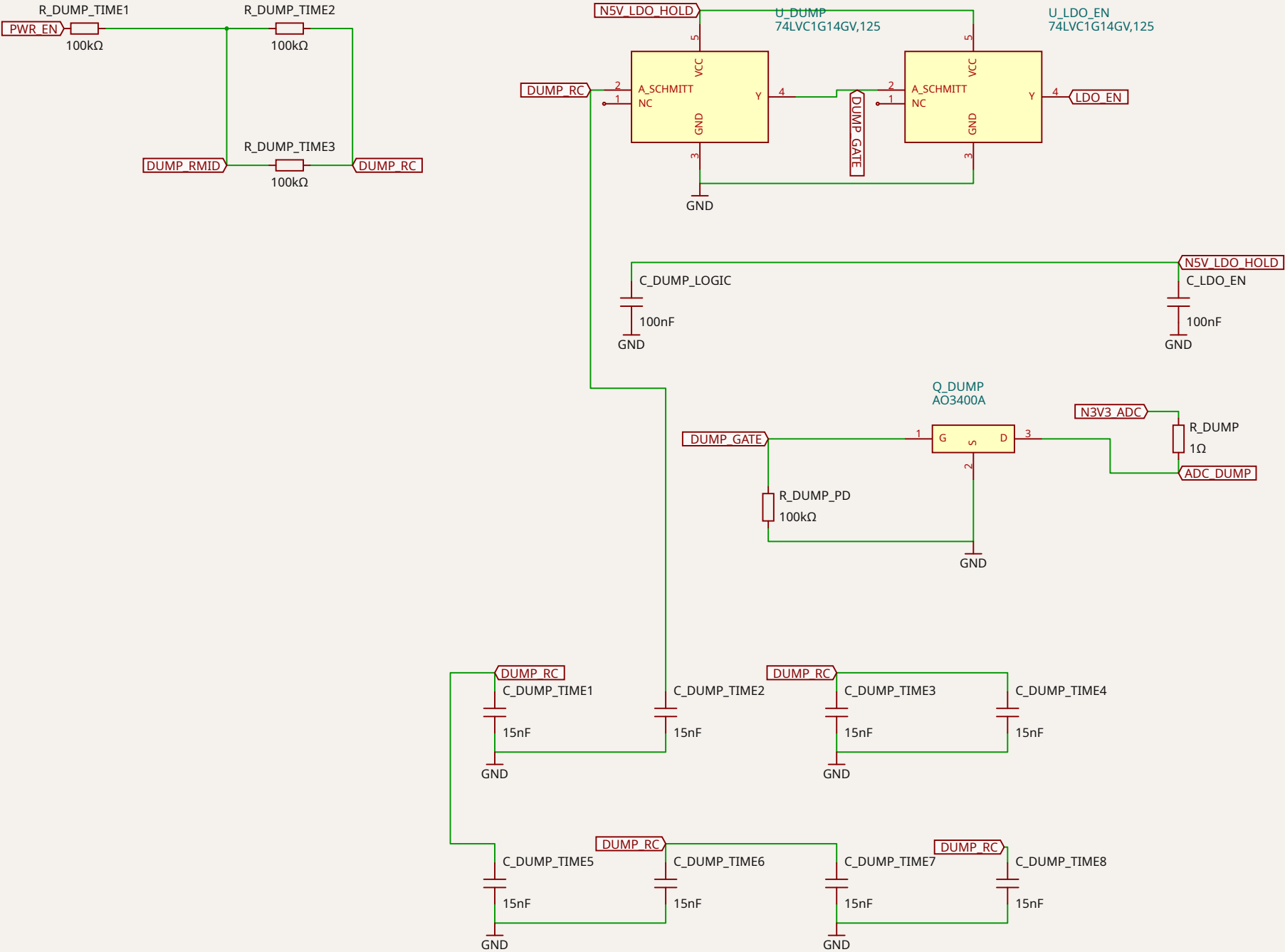
SUPERVISORS



Crow USB Carrier v1

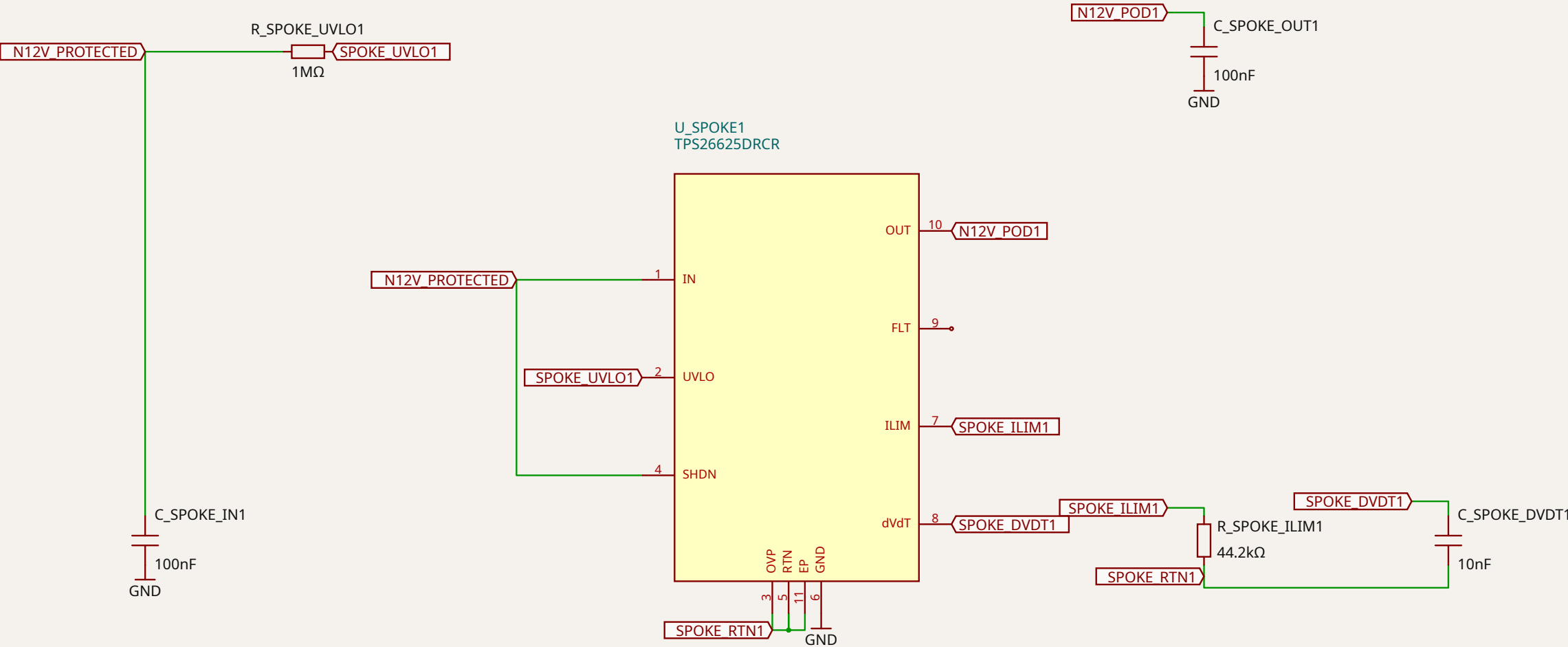
DUMP

Page 6 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 18 components • landscape page fit



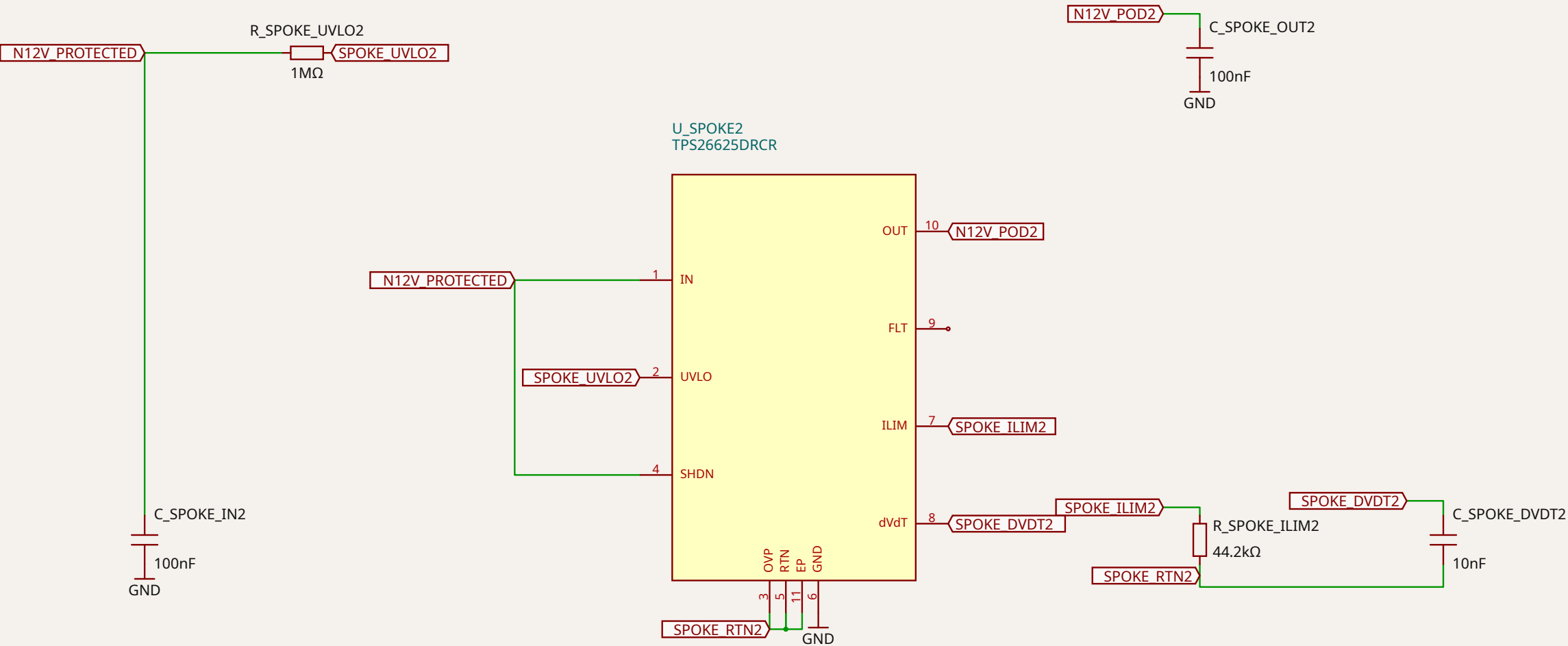
Crow USB Carrier v1

SPOKE PROTECTION 1



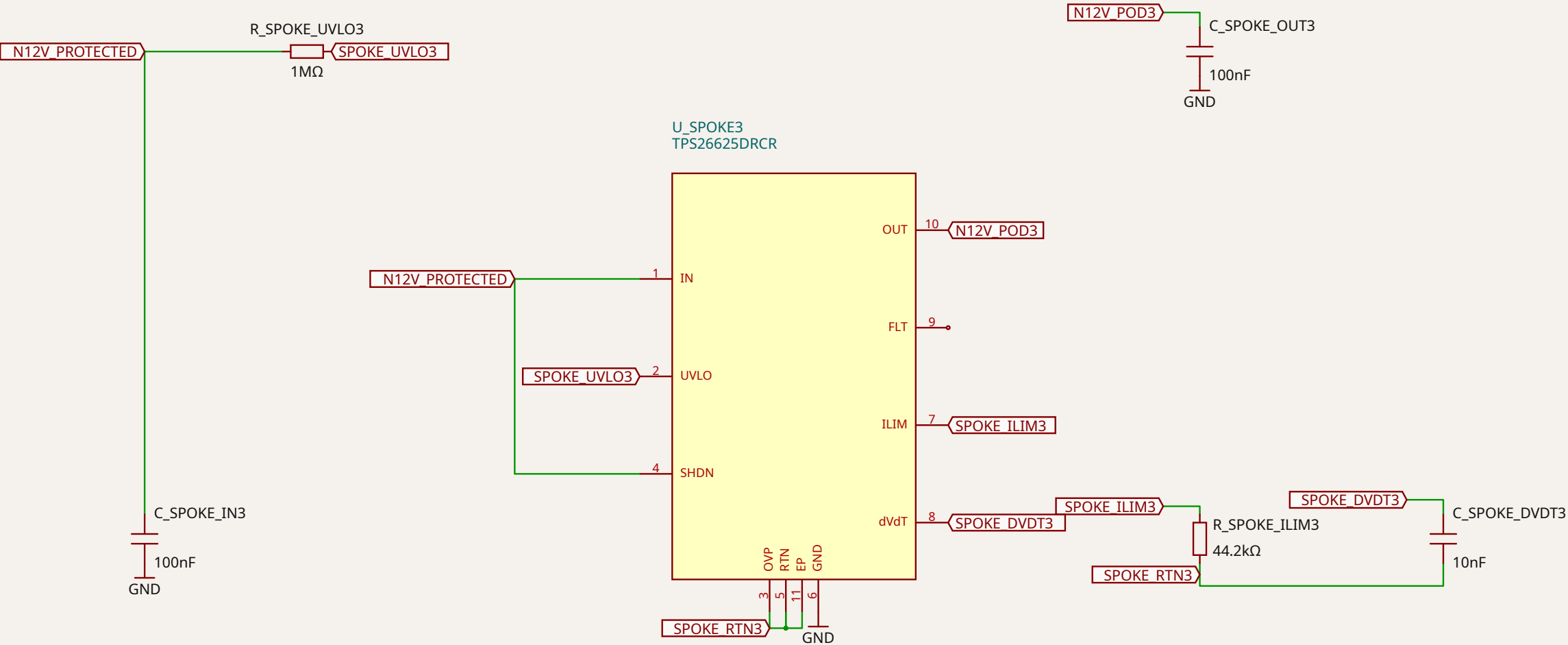
Crow USB Carrier v1

SPOKE PROTECTION 2



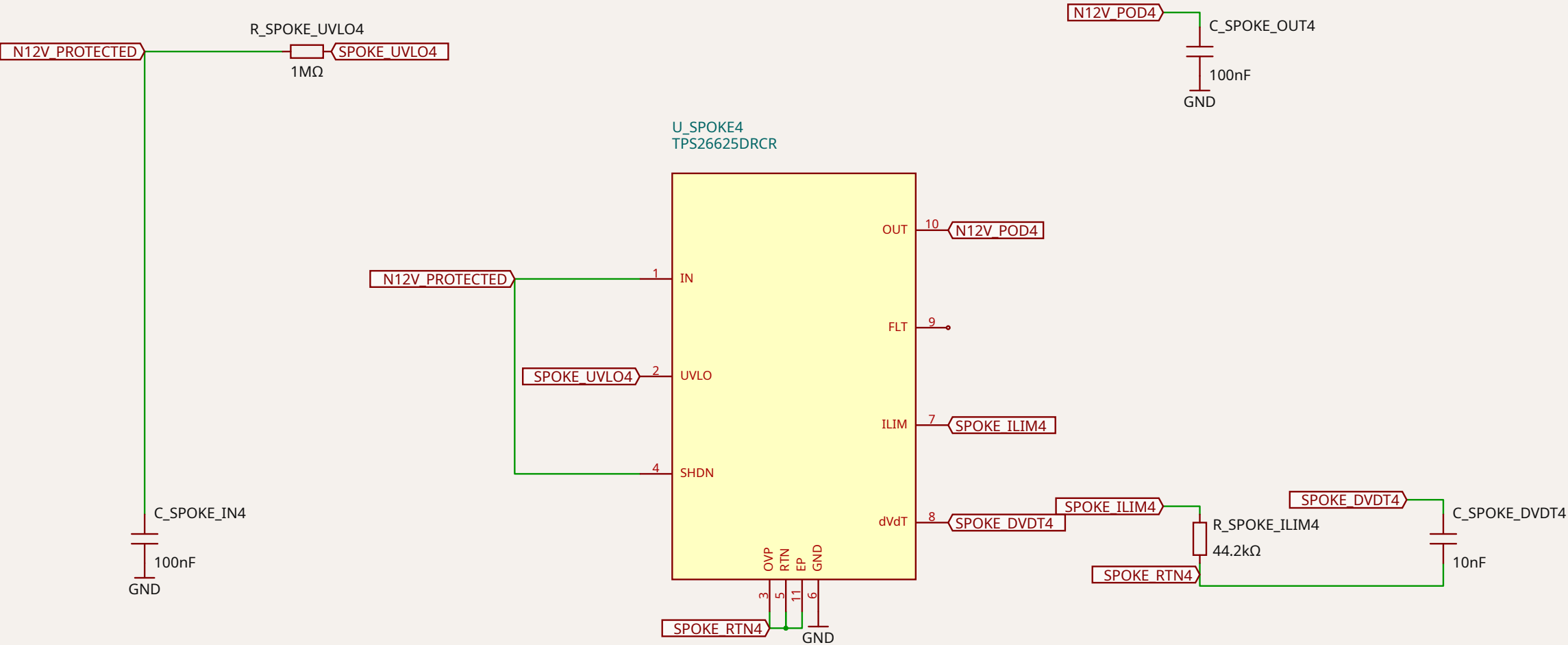
Crow USB Carrier v1

SPOKE PROTECTION 3



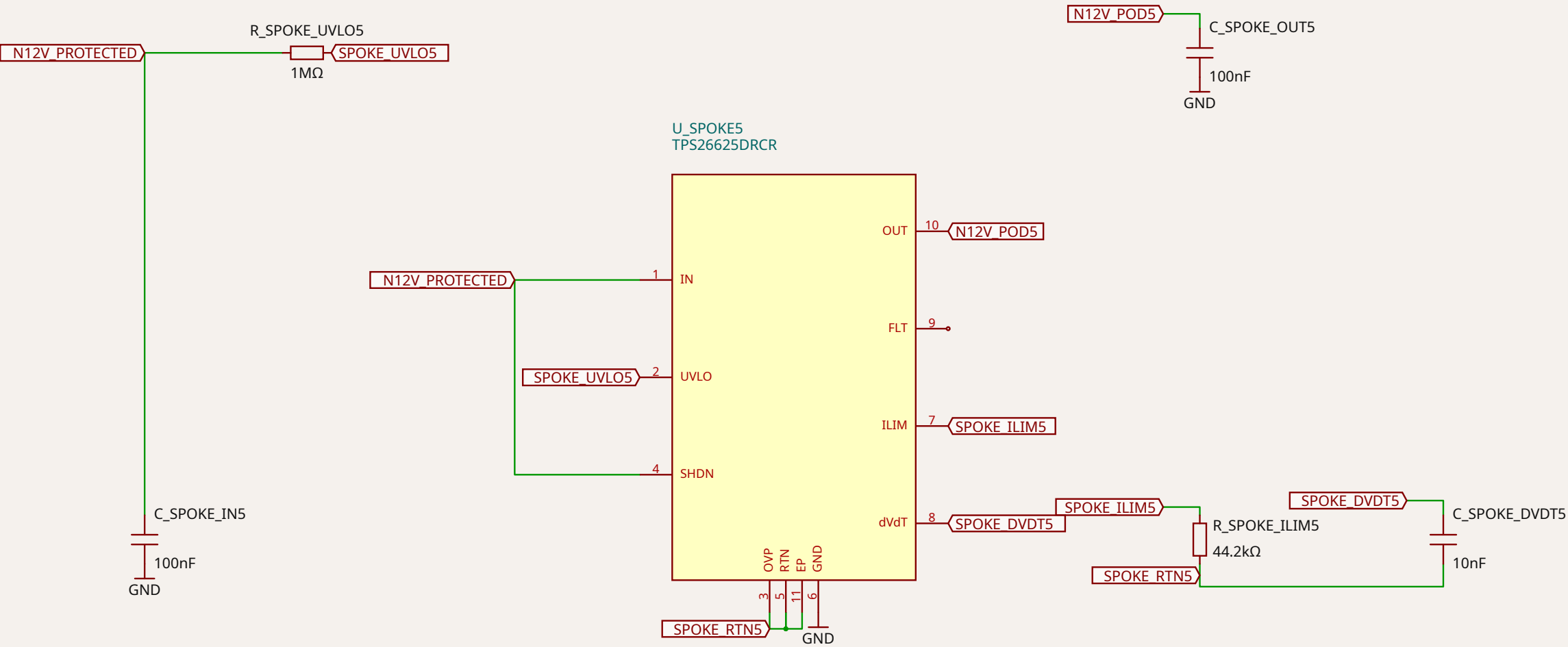
Crow USB Carrier v1

SPOKE PROTECTION 4



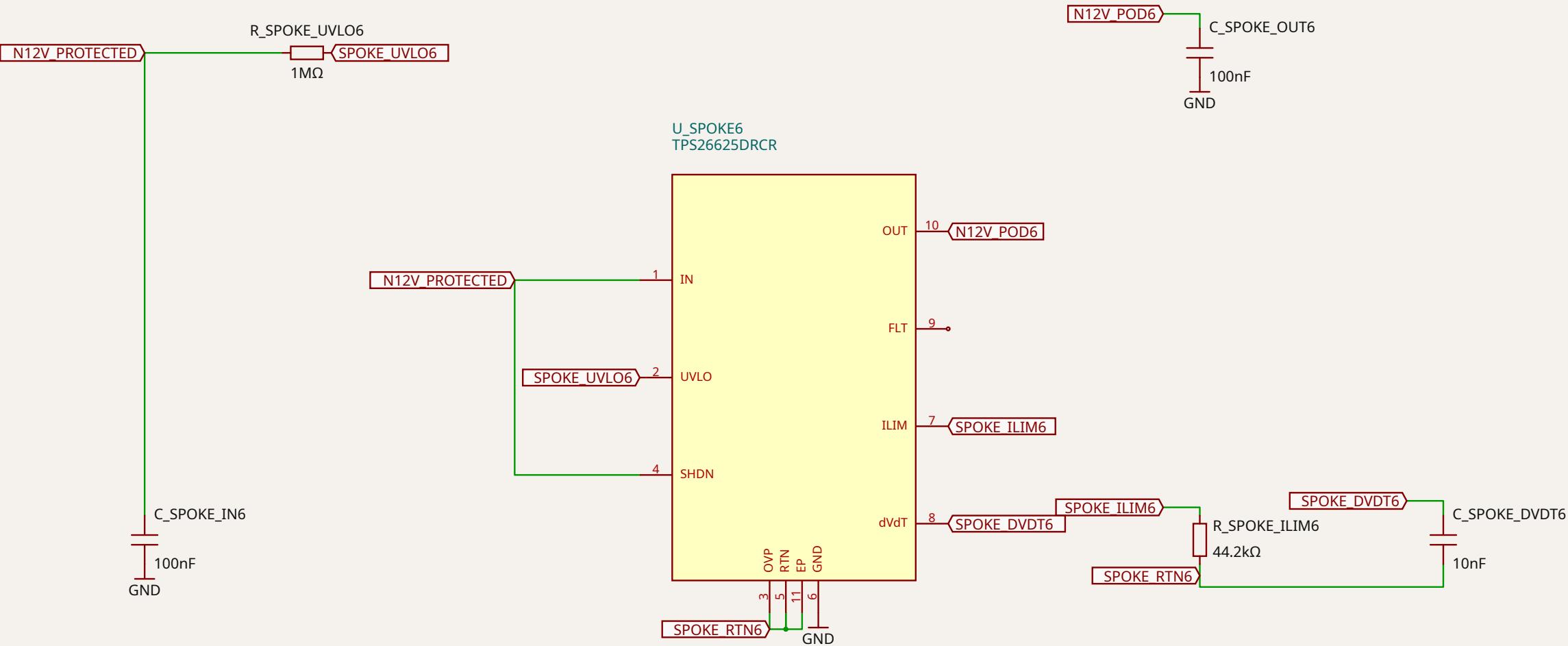
Crow USB Carrier v1

SPOKE PROTECTION 5



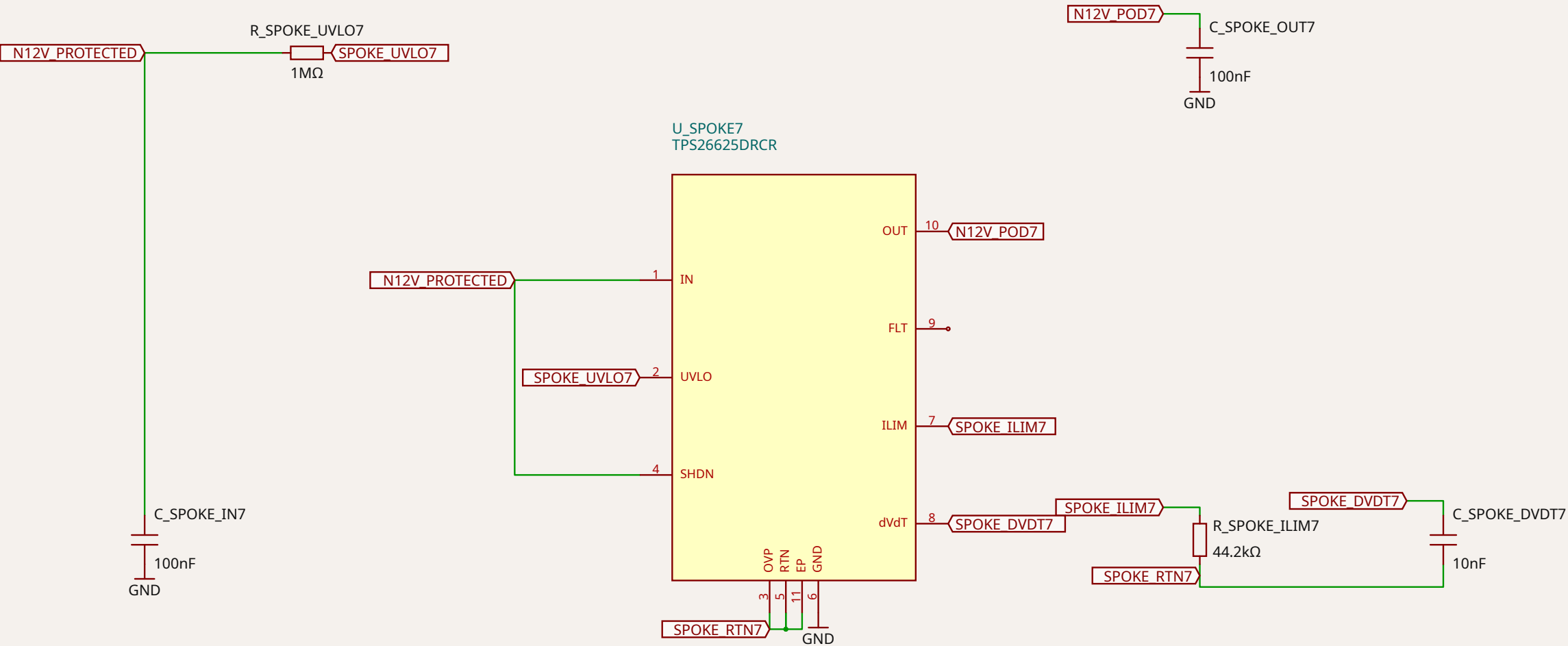
Crow USB Carrier v1

SPOKE PROTECTION 6



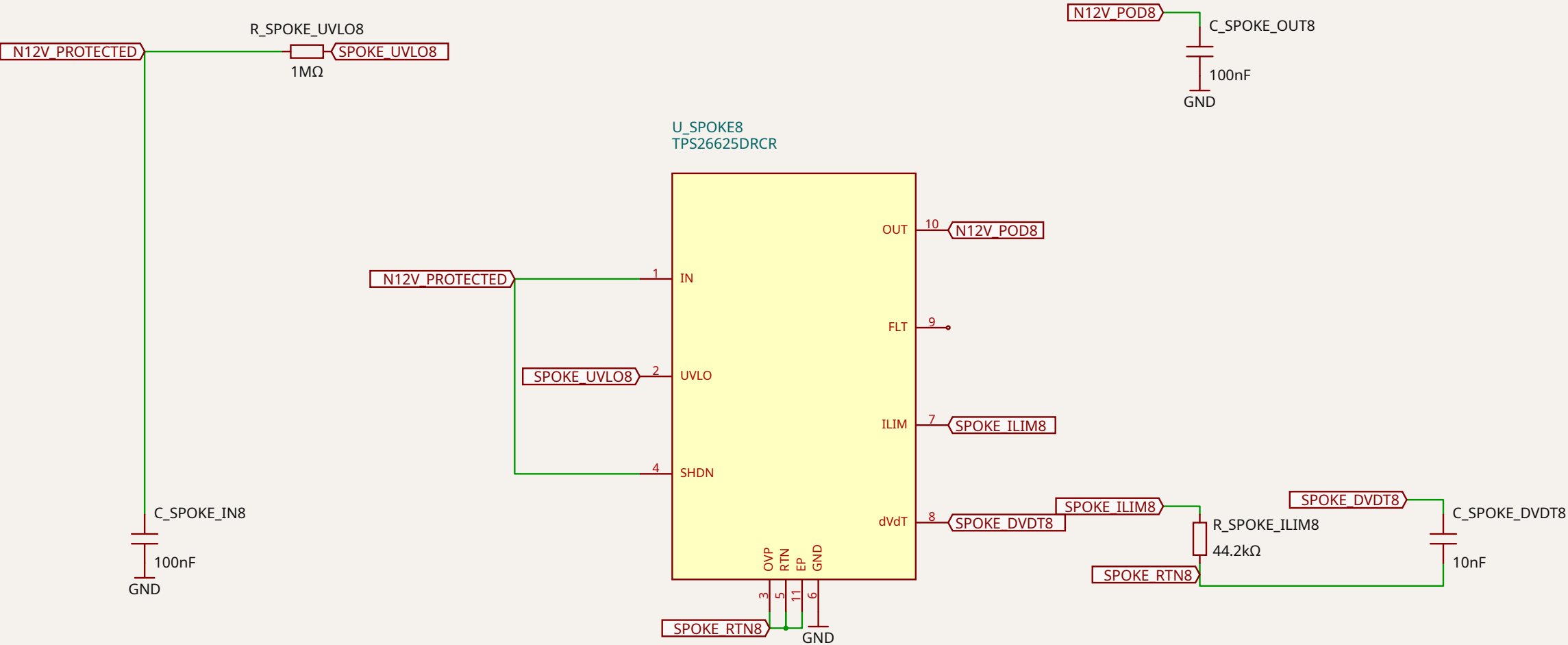
Crow USB Carrier v1

SPOKE PROTECTION 7



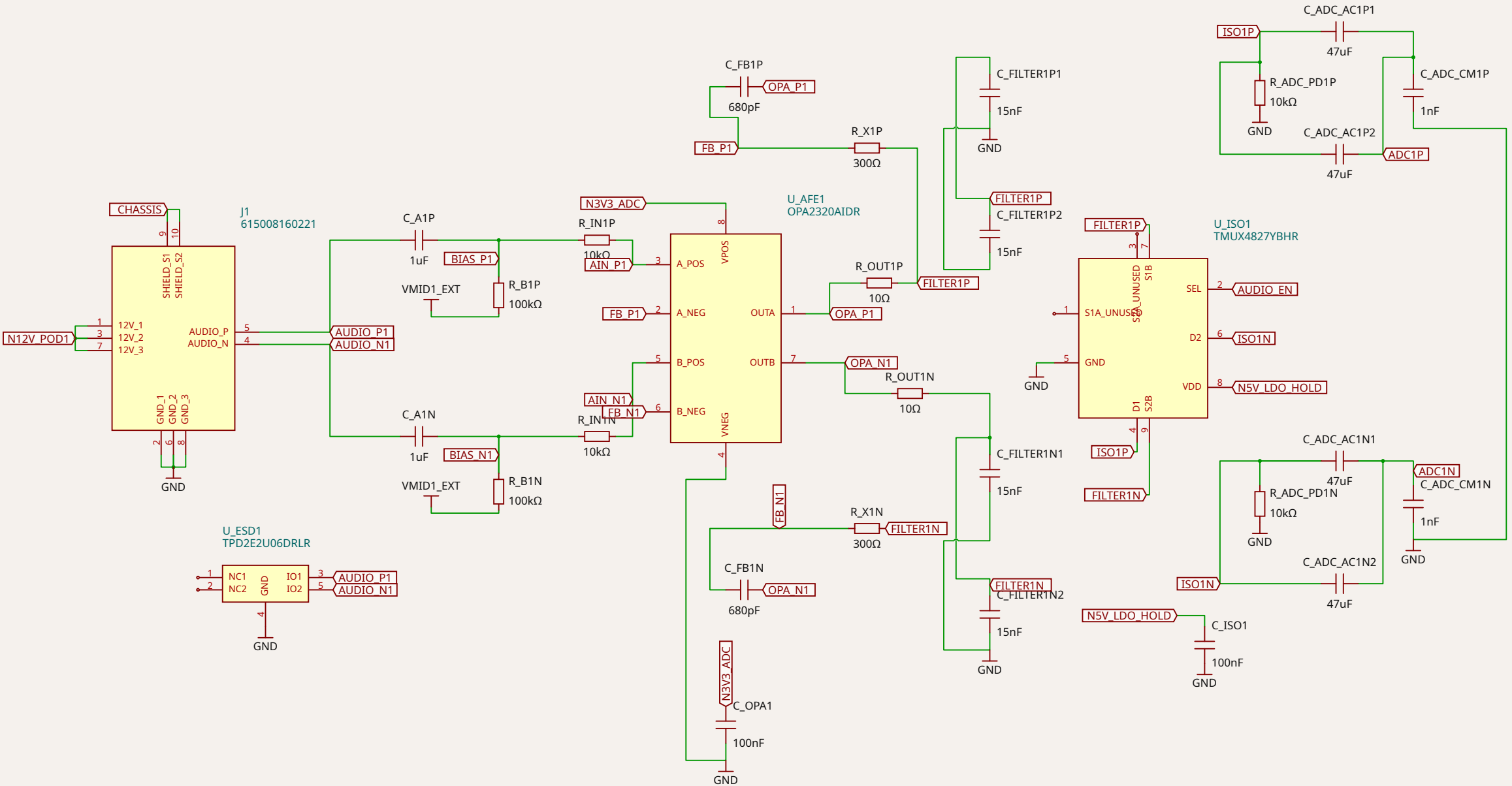
Crow USB Carrier v1

SPOKE PROTECTION 8



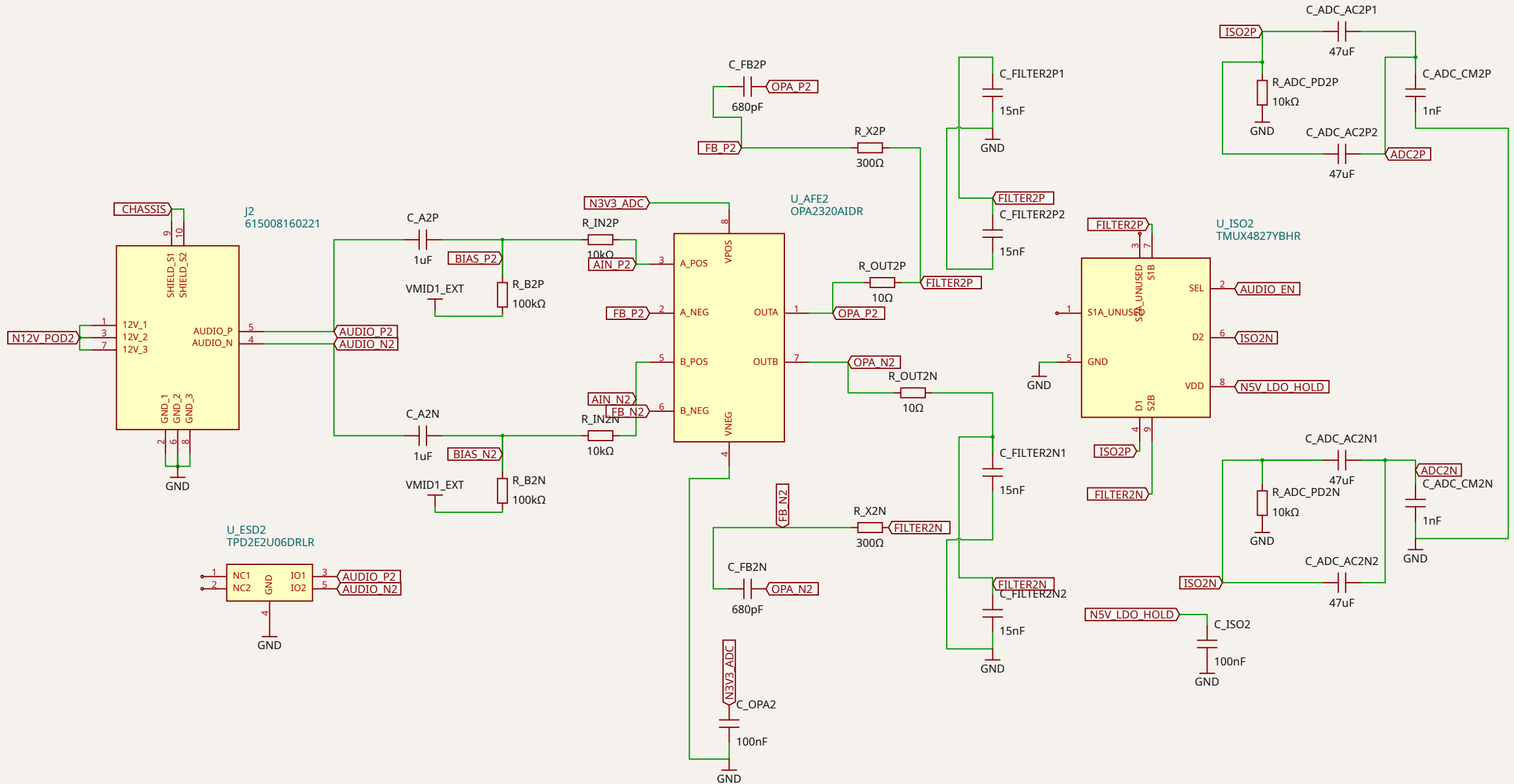
Crow USB Carrier v1

ANALOG 1



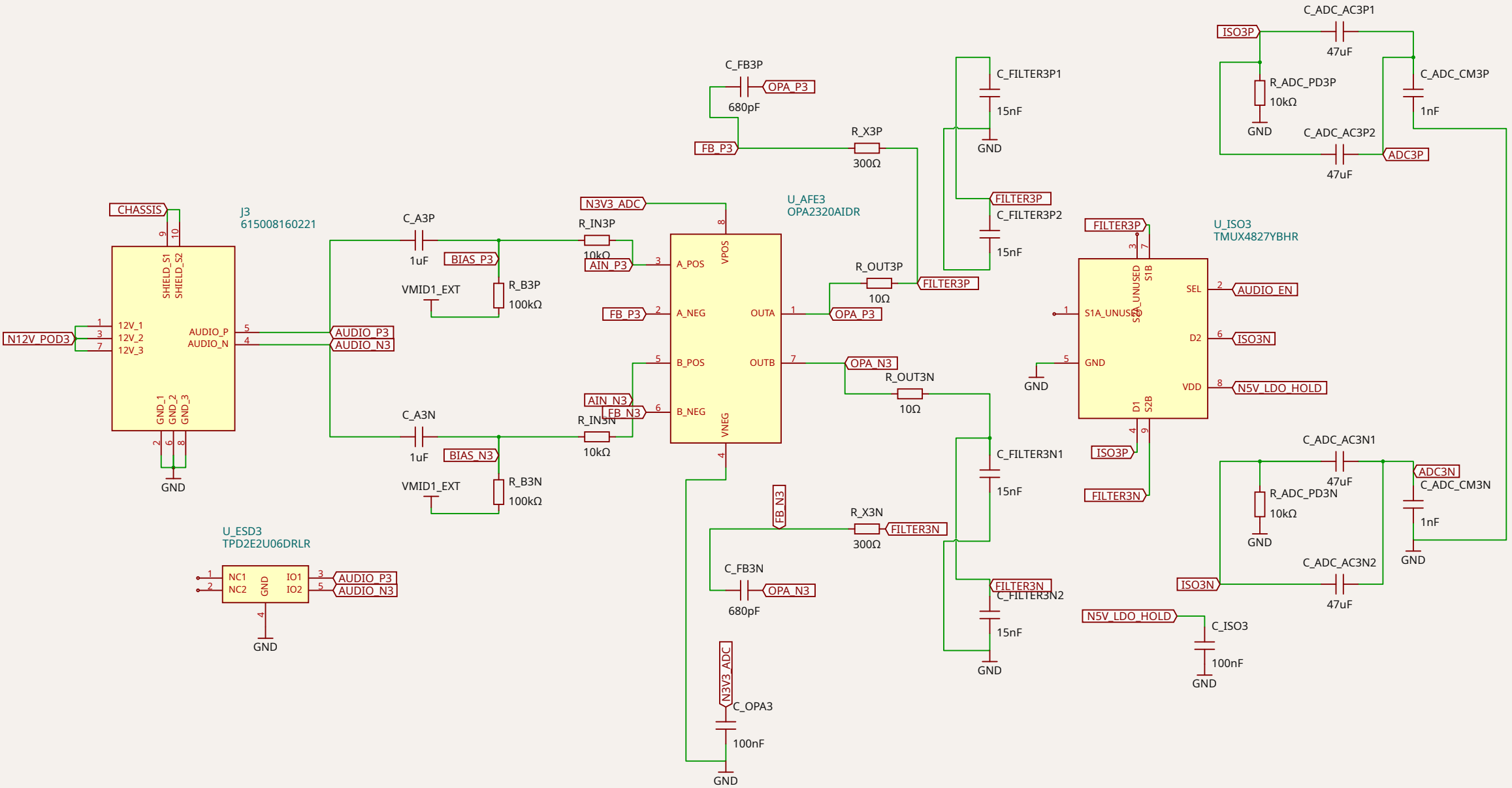
ANALOG 2

Page 16 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 30 components • landscape page fit



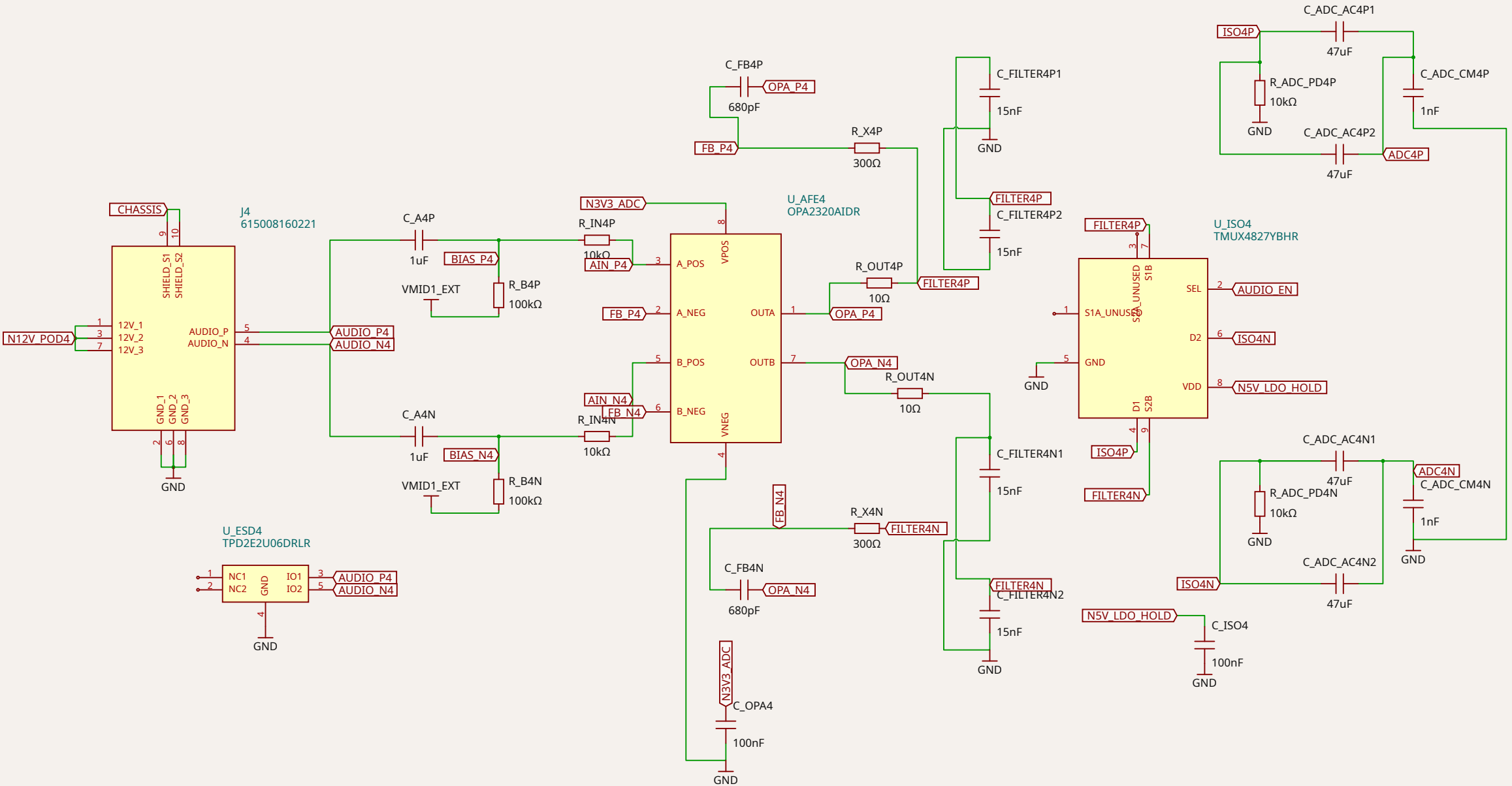
Crow USB Carrier v1

ANALOG 3



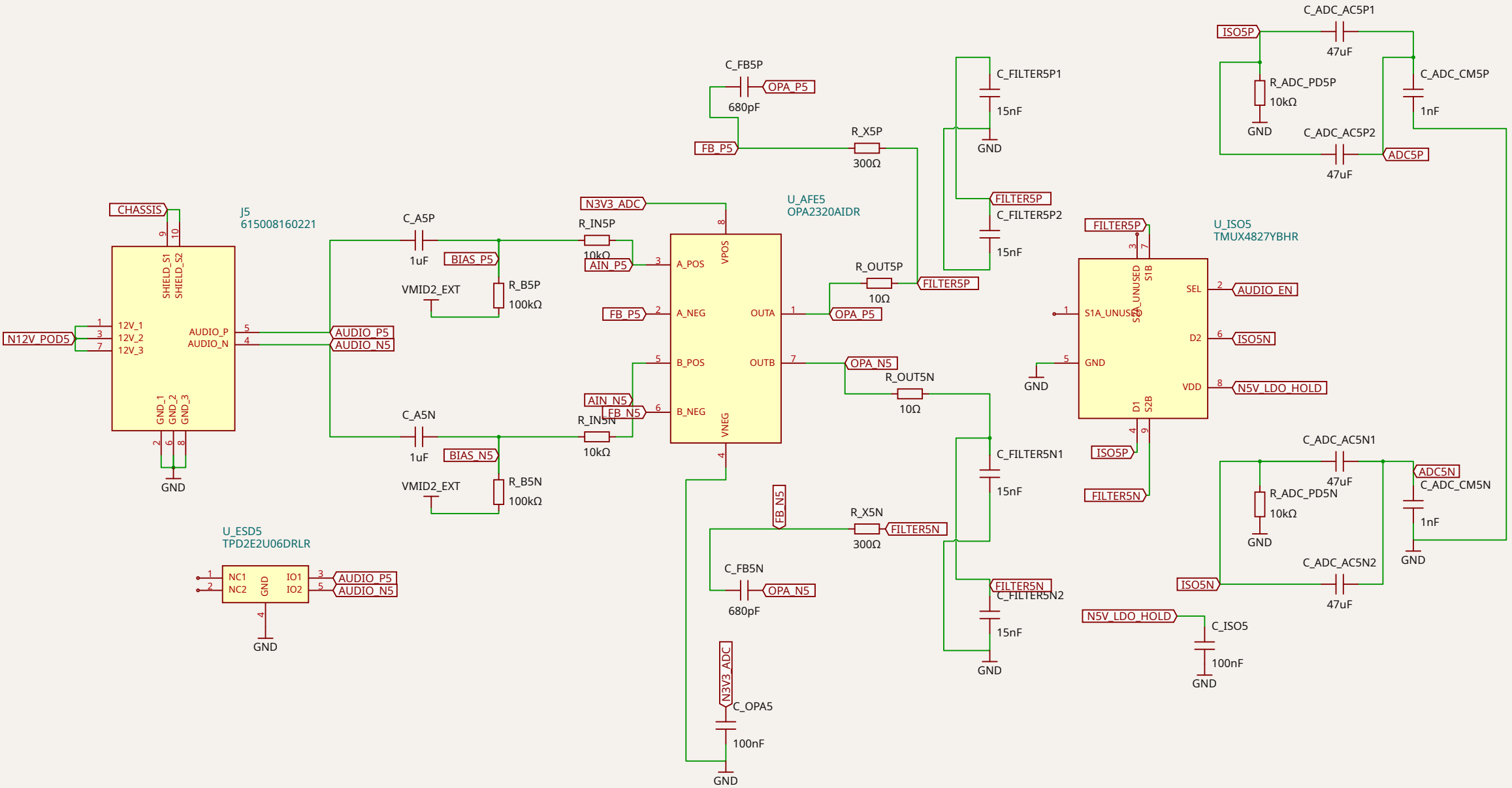
Crow USB Carrier v1

ANALOG 4



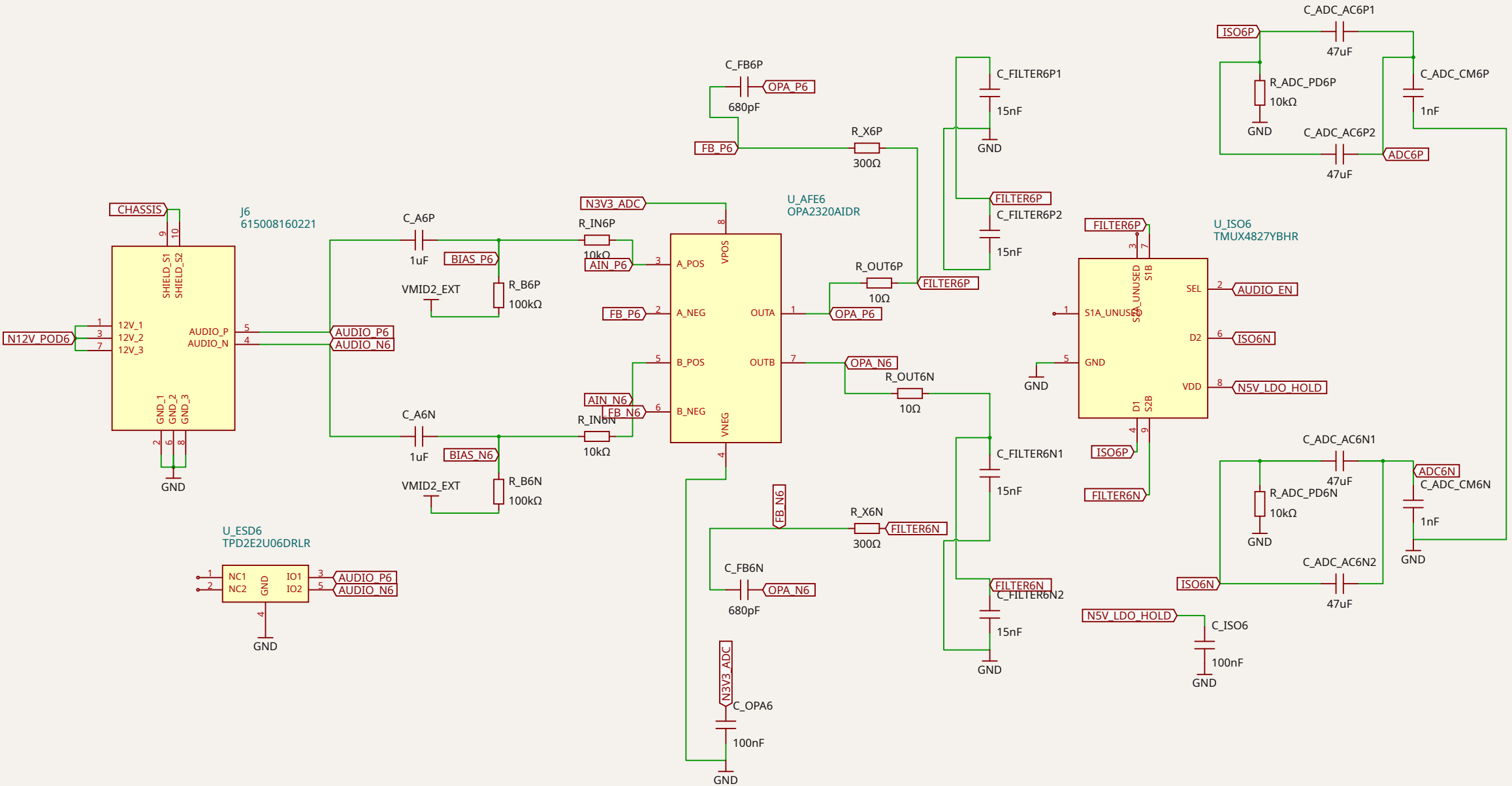
Crow USB Carrier v1

ANALOG 5



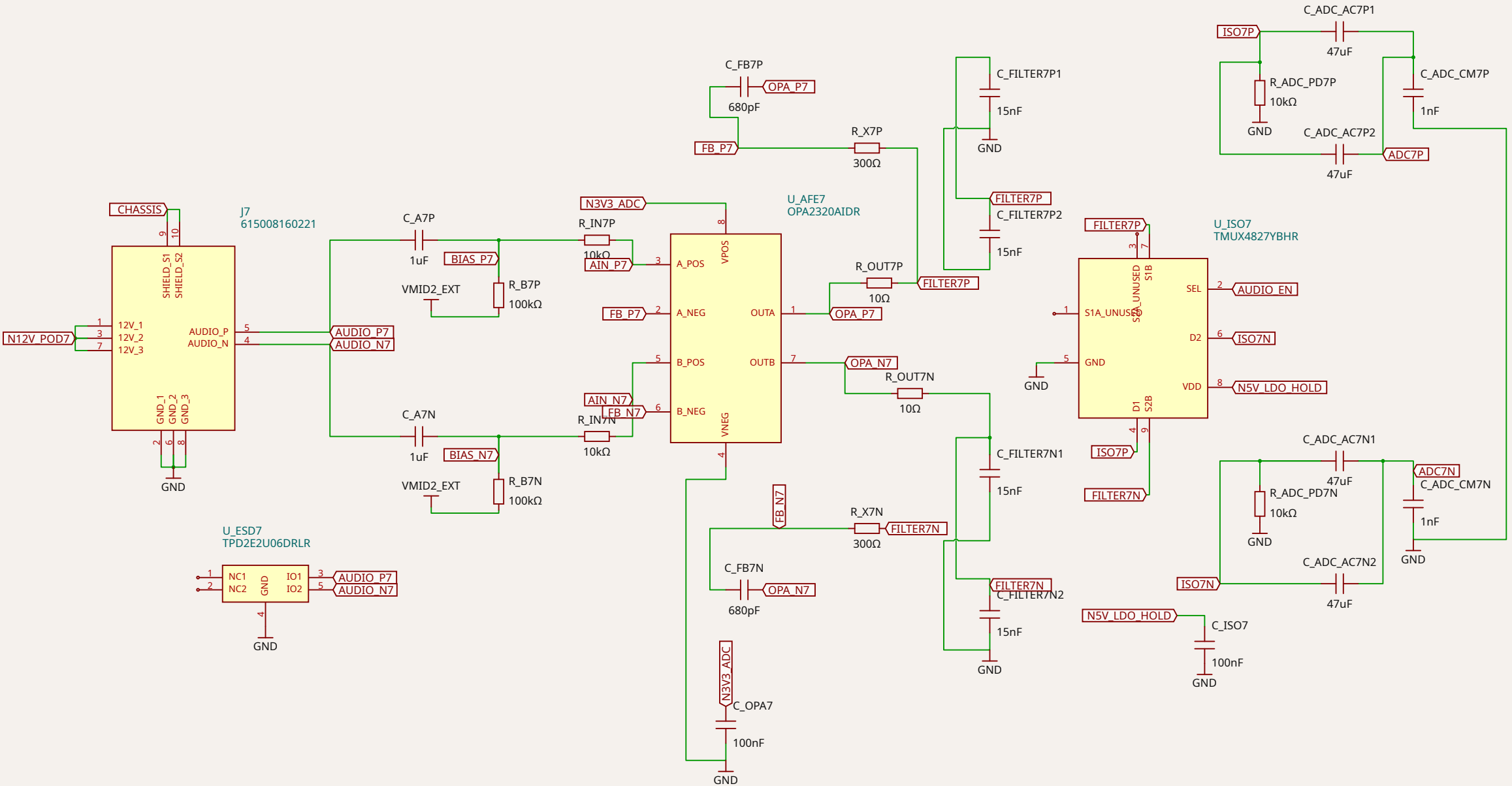
Crow USB Carrier v1

ANALOG 6



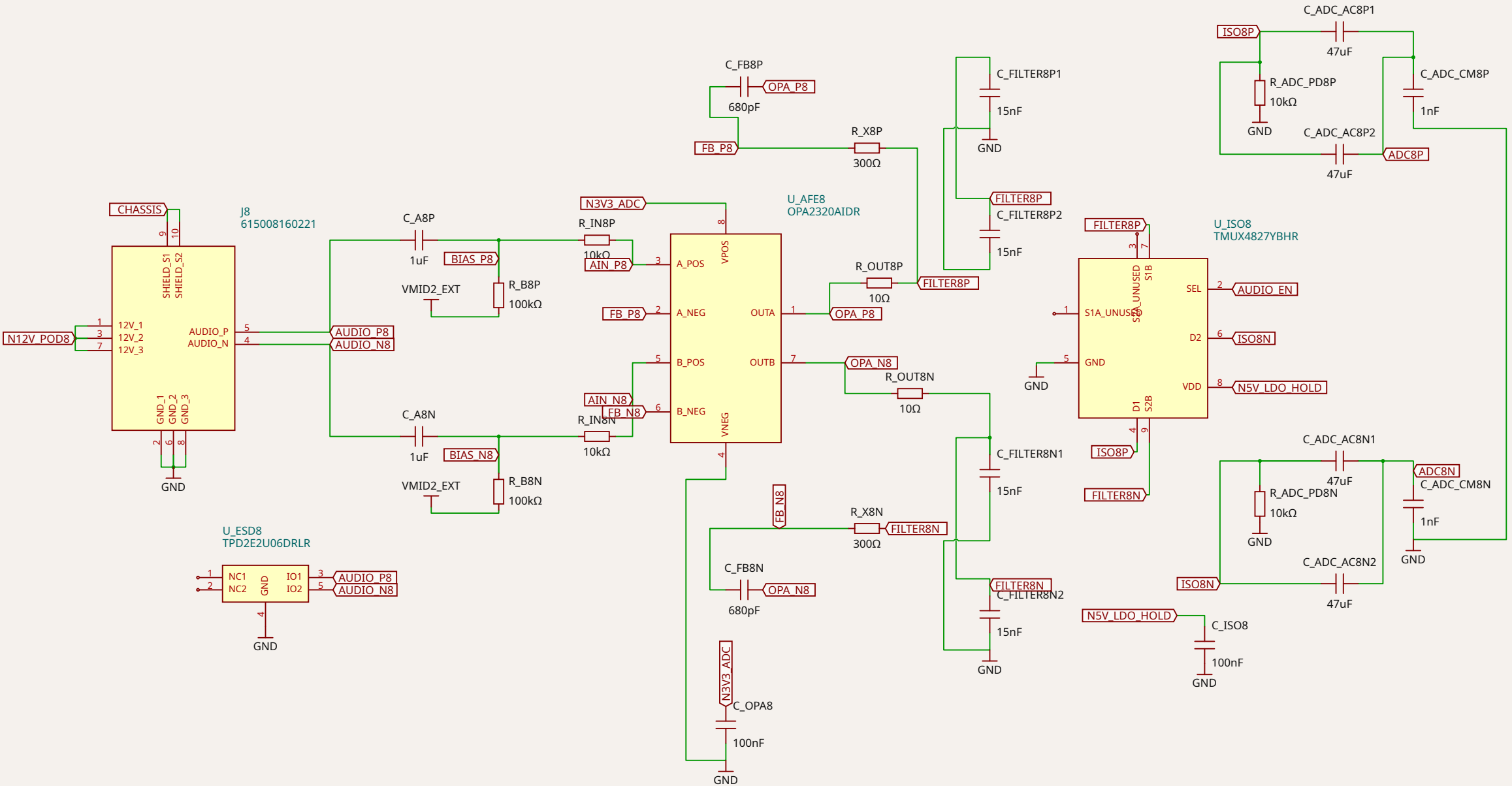
Crow USB Carrier v1

ANALOG 7



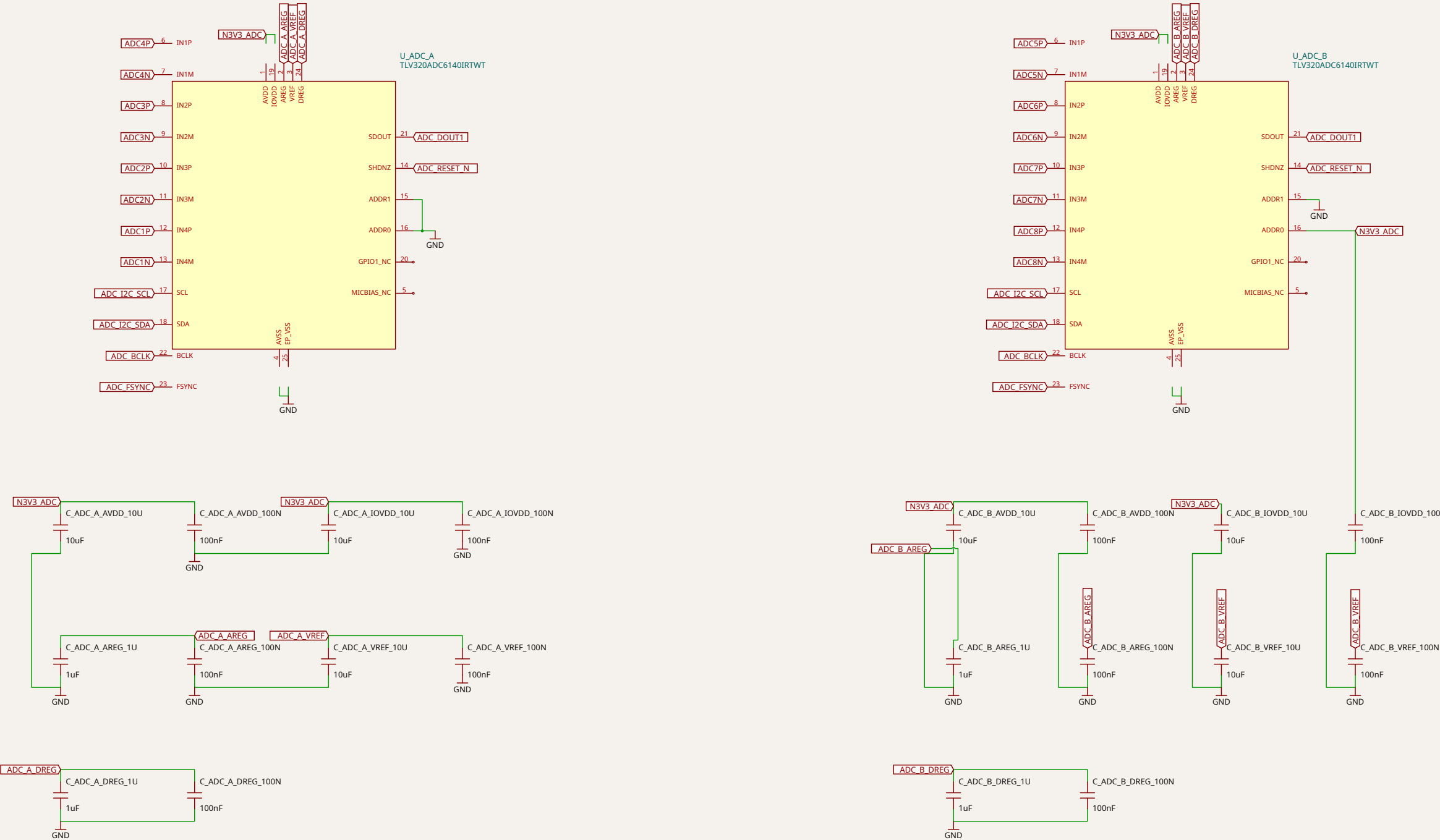
Crow USB Carrier v1

ANALOG 8



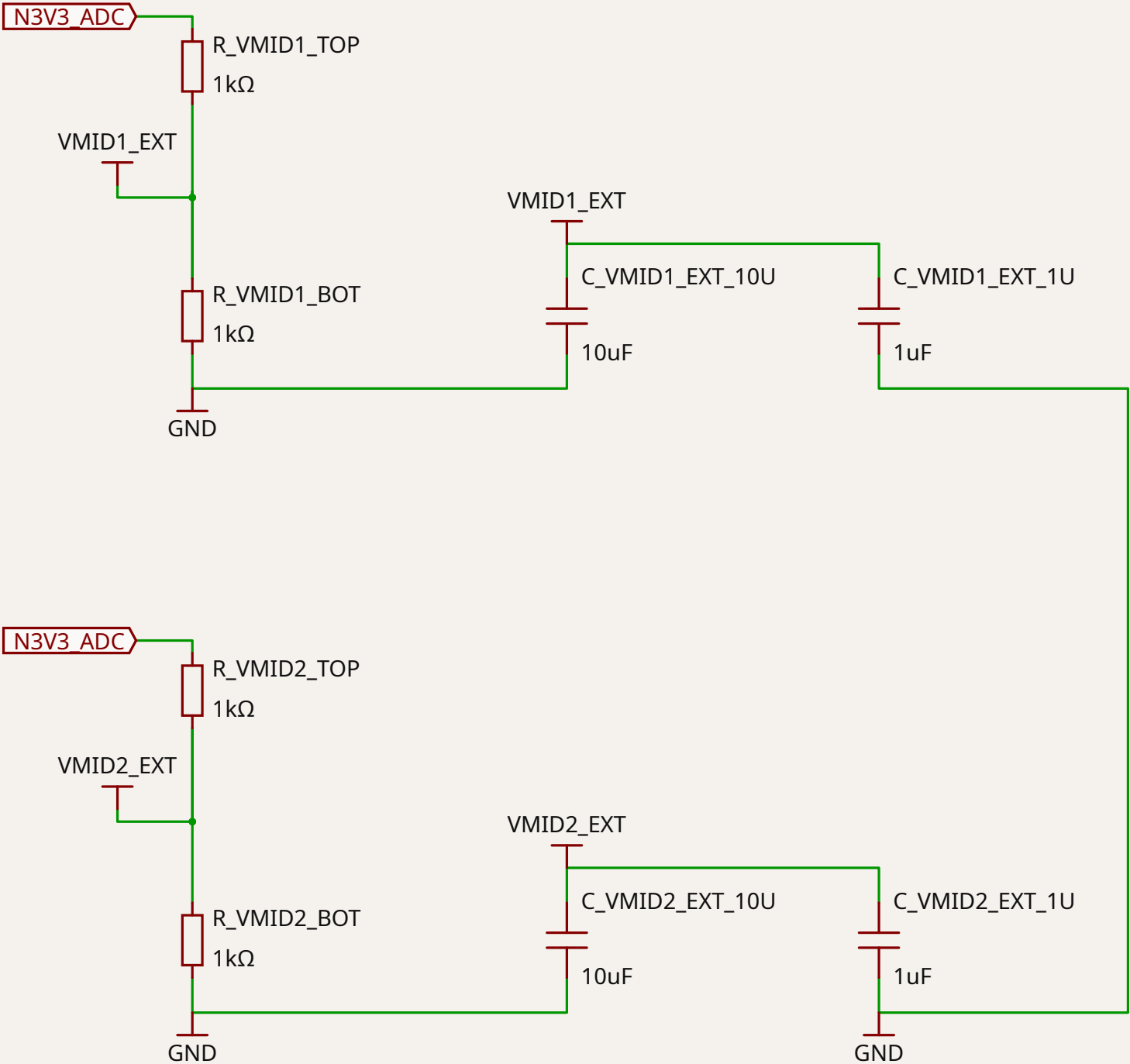
Crow USB Carrier v1

ADC



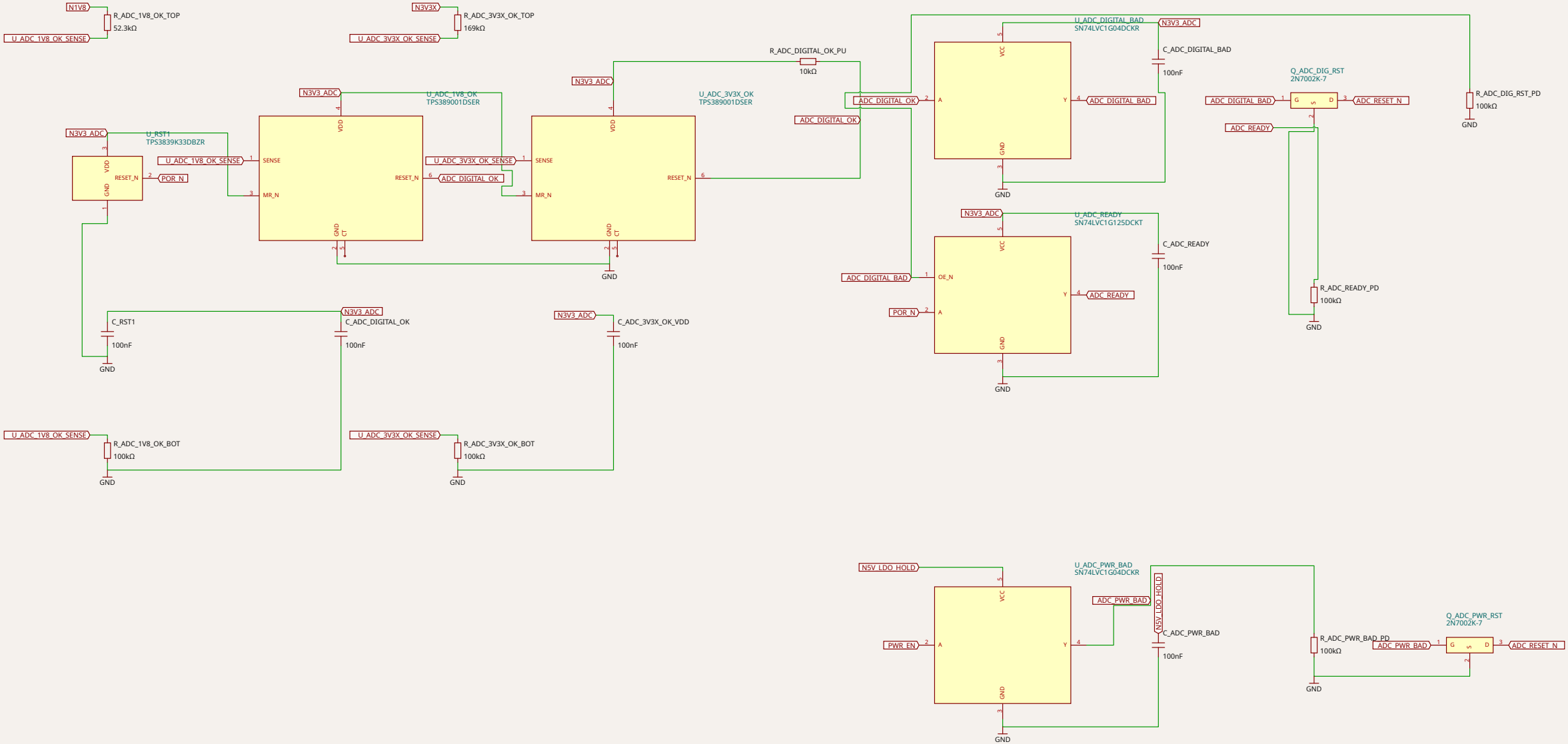
Crow USB Carrier v1

VMID



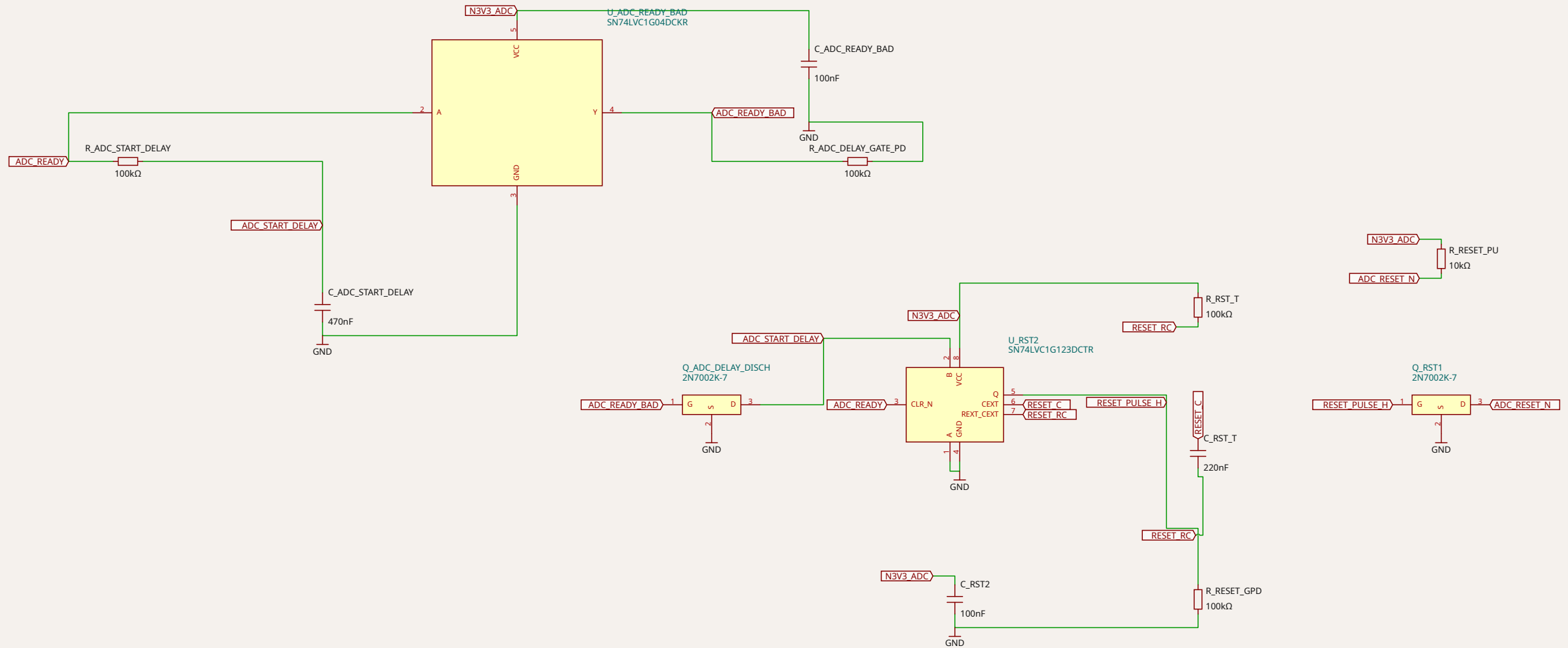
Crow USB Carrier v1

RESET SUPERVISORS



Crow USB Carrier v1

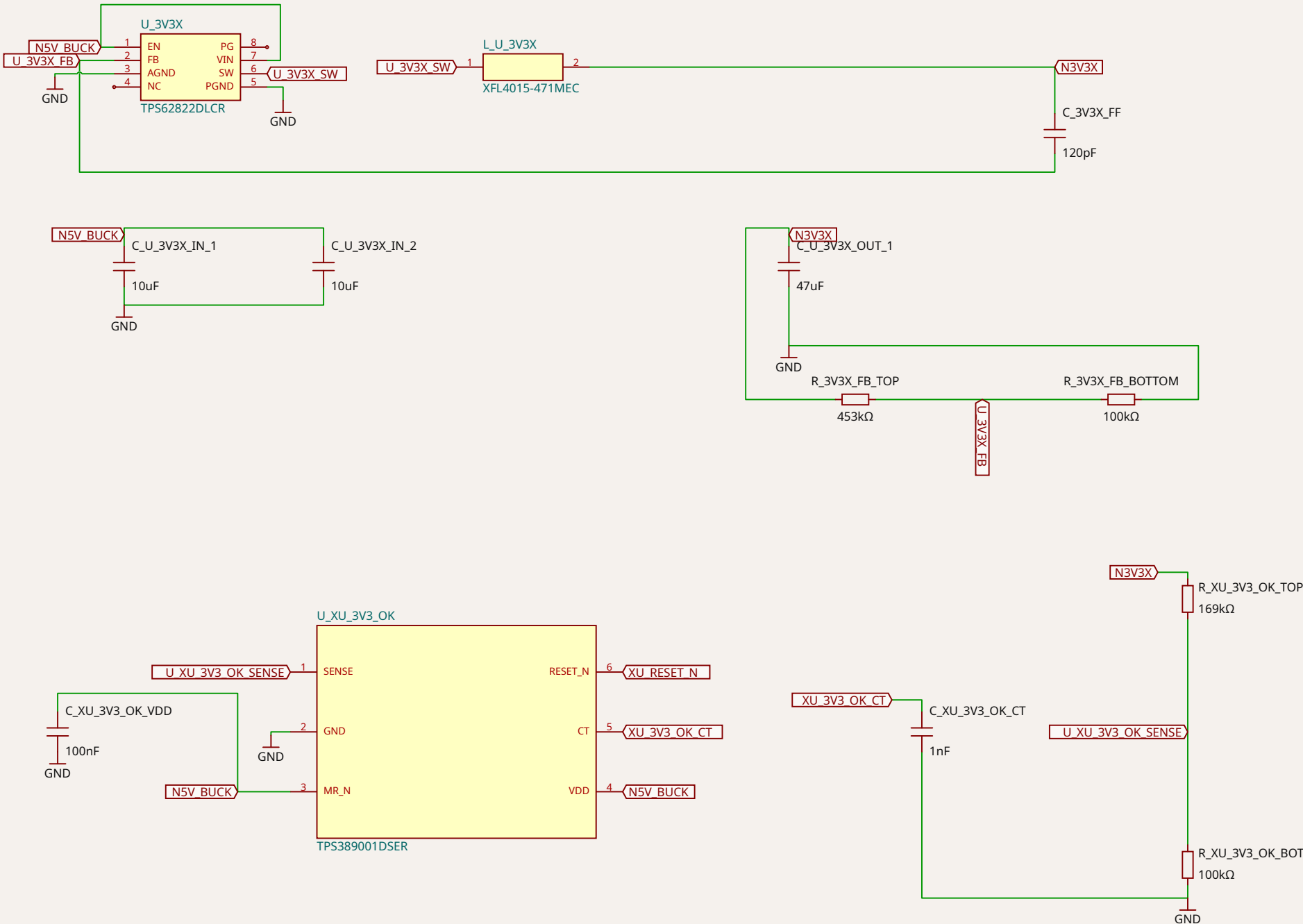
RESET SEQUENCER



Crow USB Carrier v1

DIGITAL POWER 3V3X

Page 27 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 13 components • landscape page fit

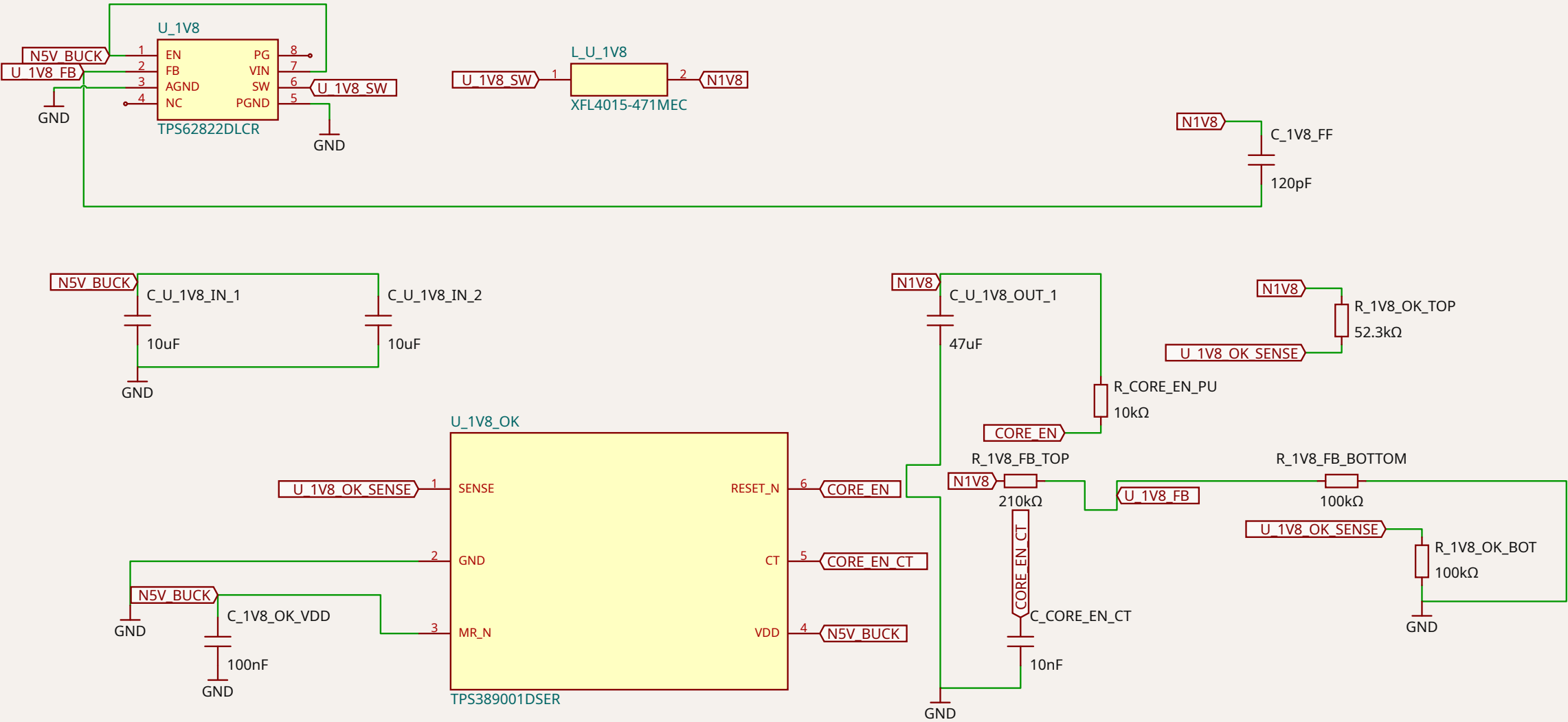


Crow USB Carrier v1

DIGITAL POWER 1V8

Page 28 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 14 components • landscape page fit

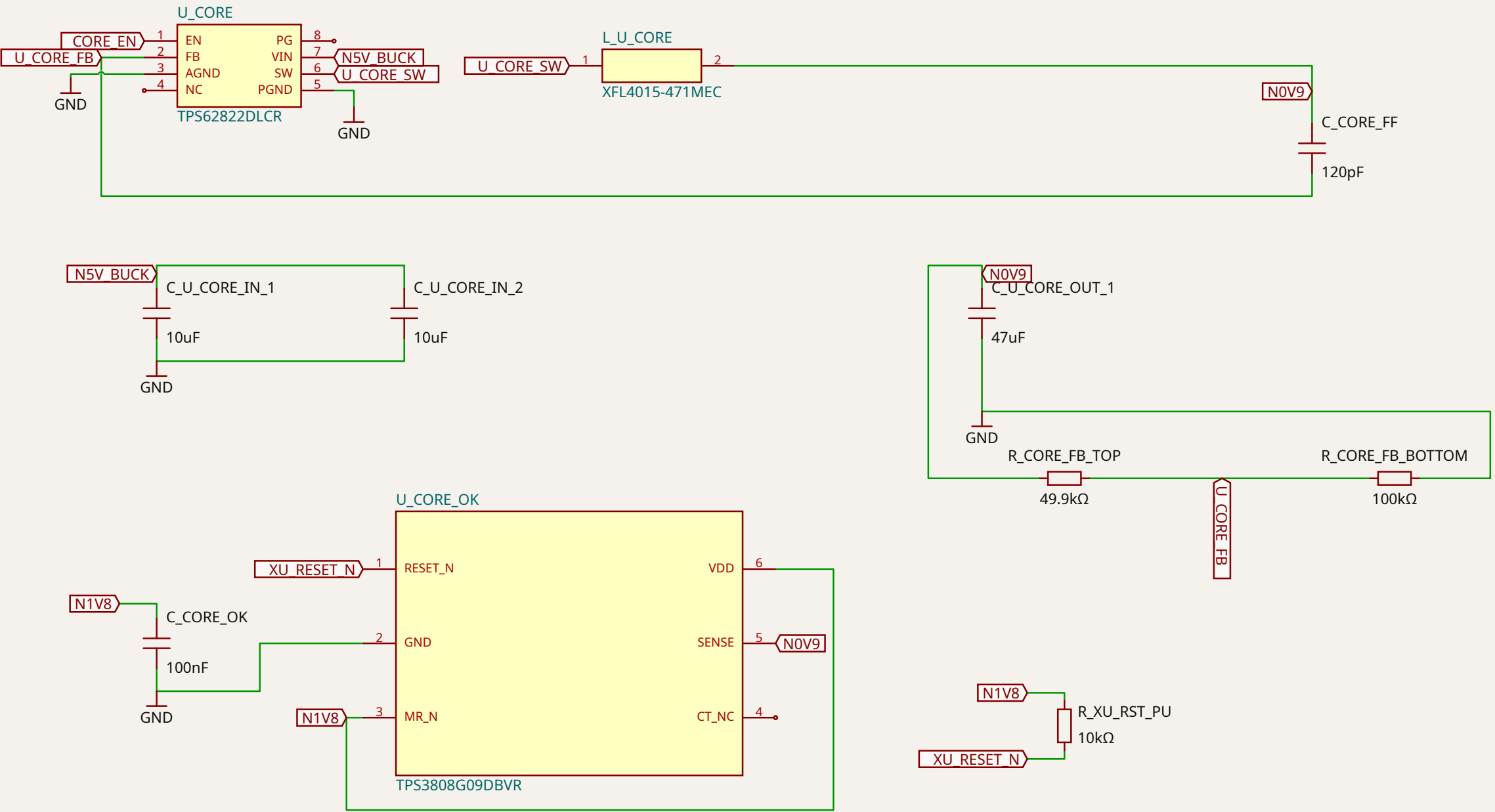
Page 28 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 14 components • landscape page fit



Crow USB Carrier v1

DIGITAL POWER CORE

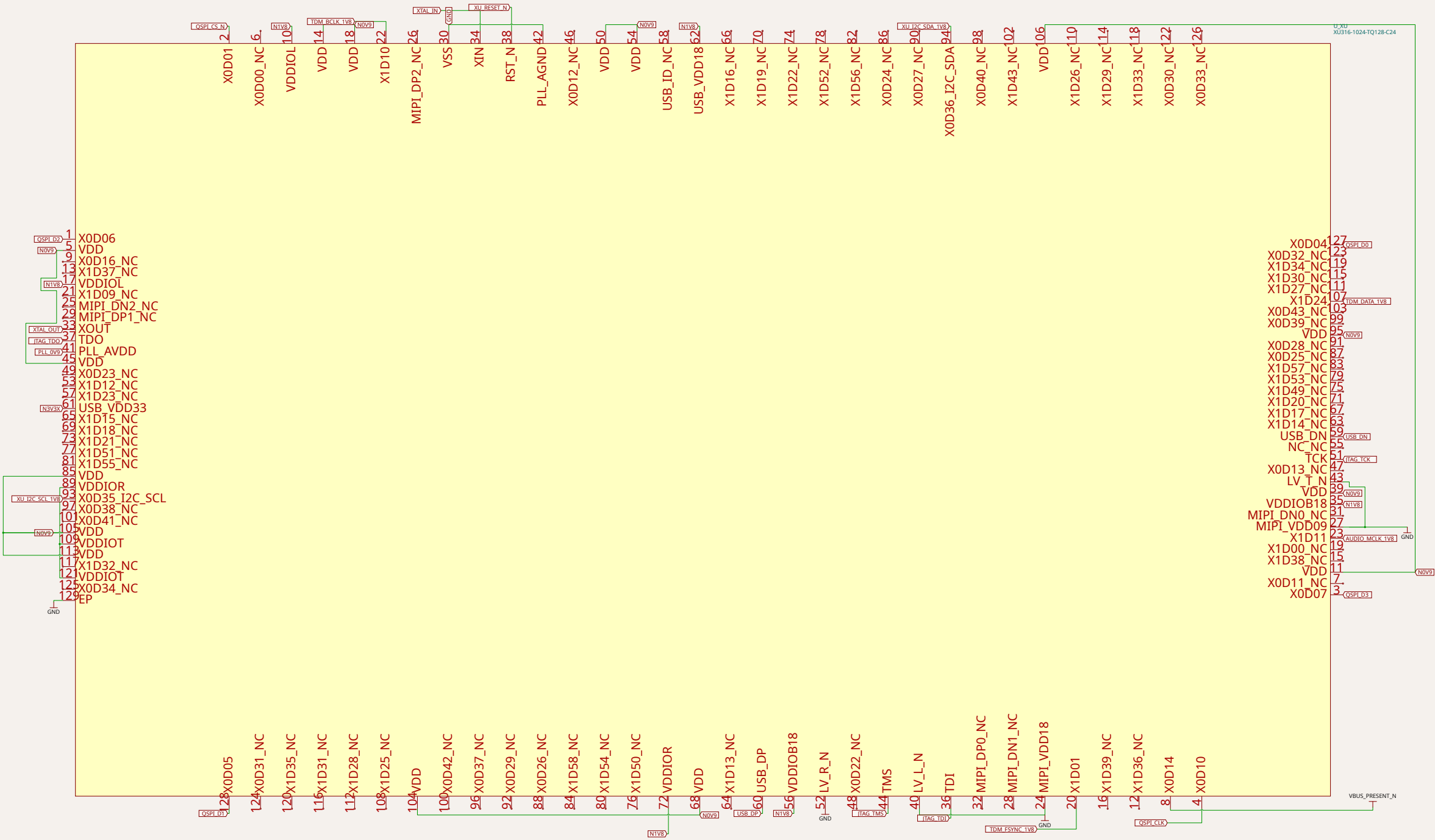
Page 29 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 11 components • landscape page fit



Crow USB Carrier v1

XMOS CORE

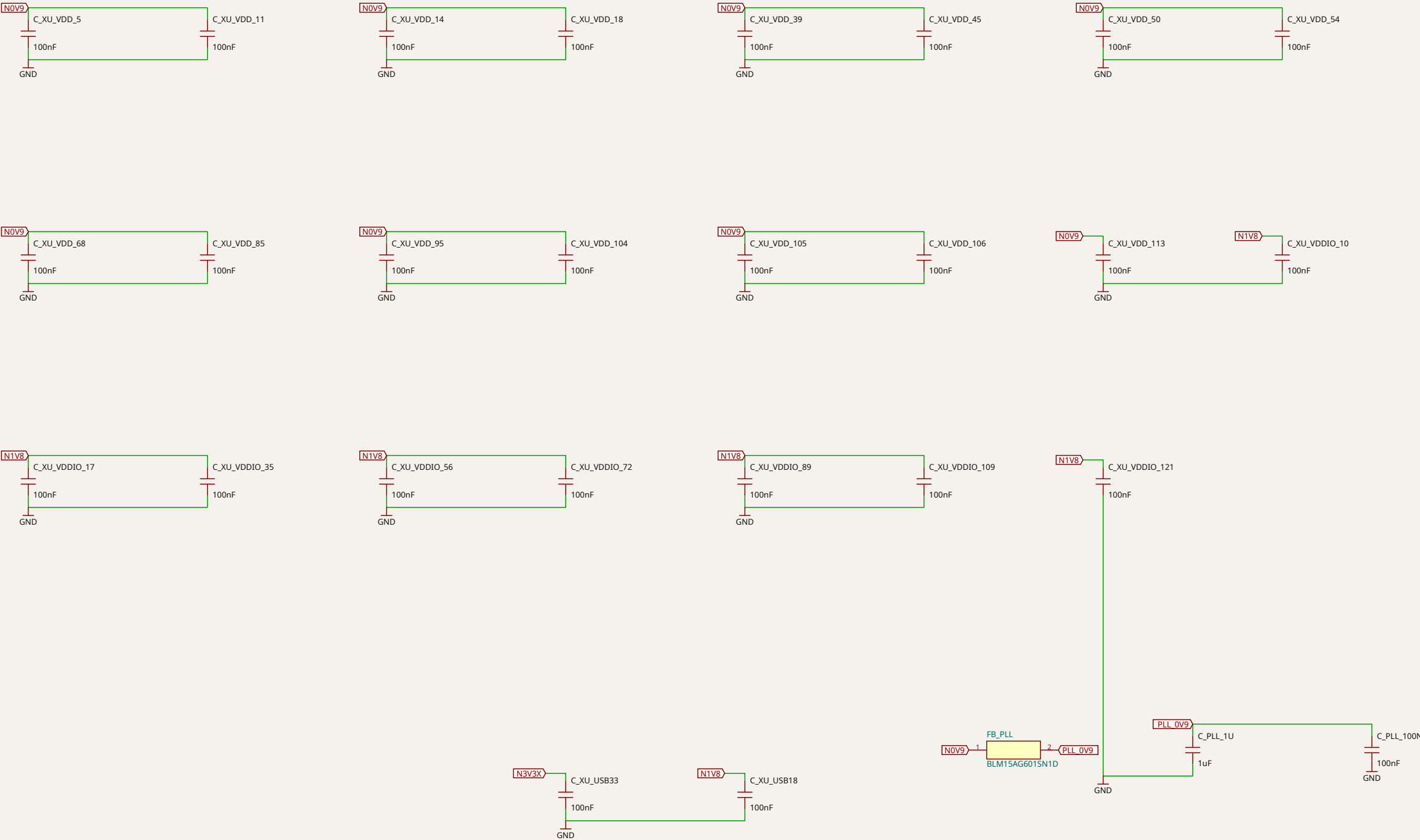
Page 30 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 1 components • landscape page fit



Crow USB Carrier v1

XMOS DECOUPLING

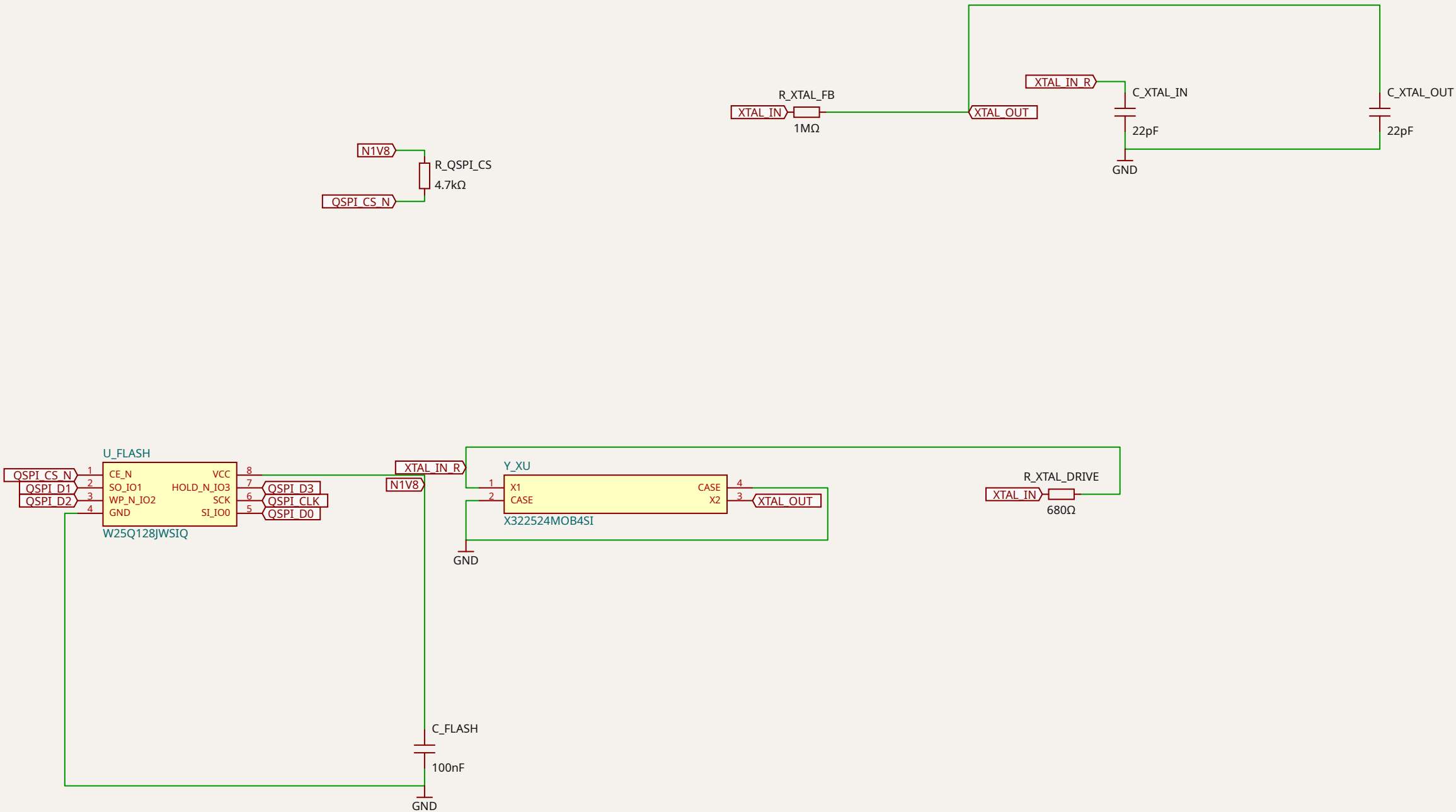
Page 31 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 28 components • landscape page fit



Crow USB Carrier v1

FLASH CLOCK

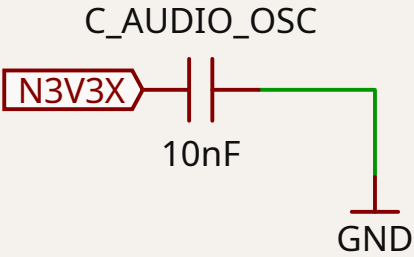
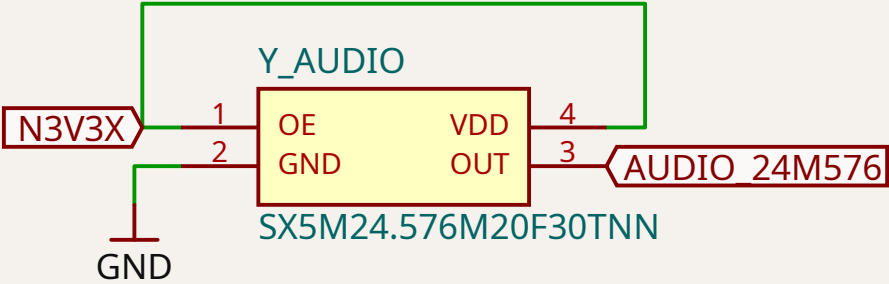
Page 32 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 8 components • landscape page fit



Crow USB Carrier v1

AUDIO OSCILLATOR

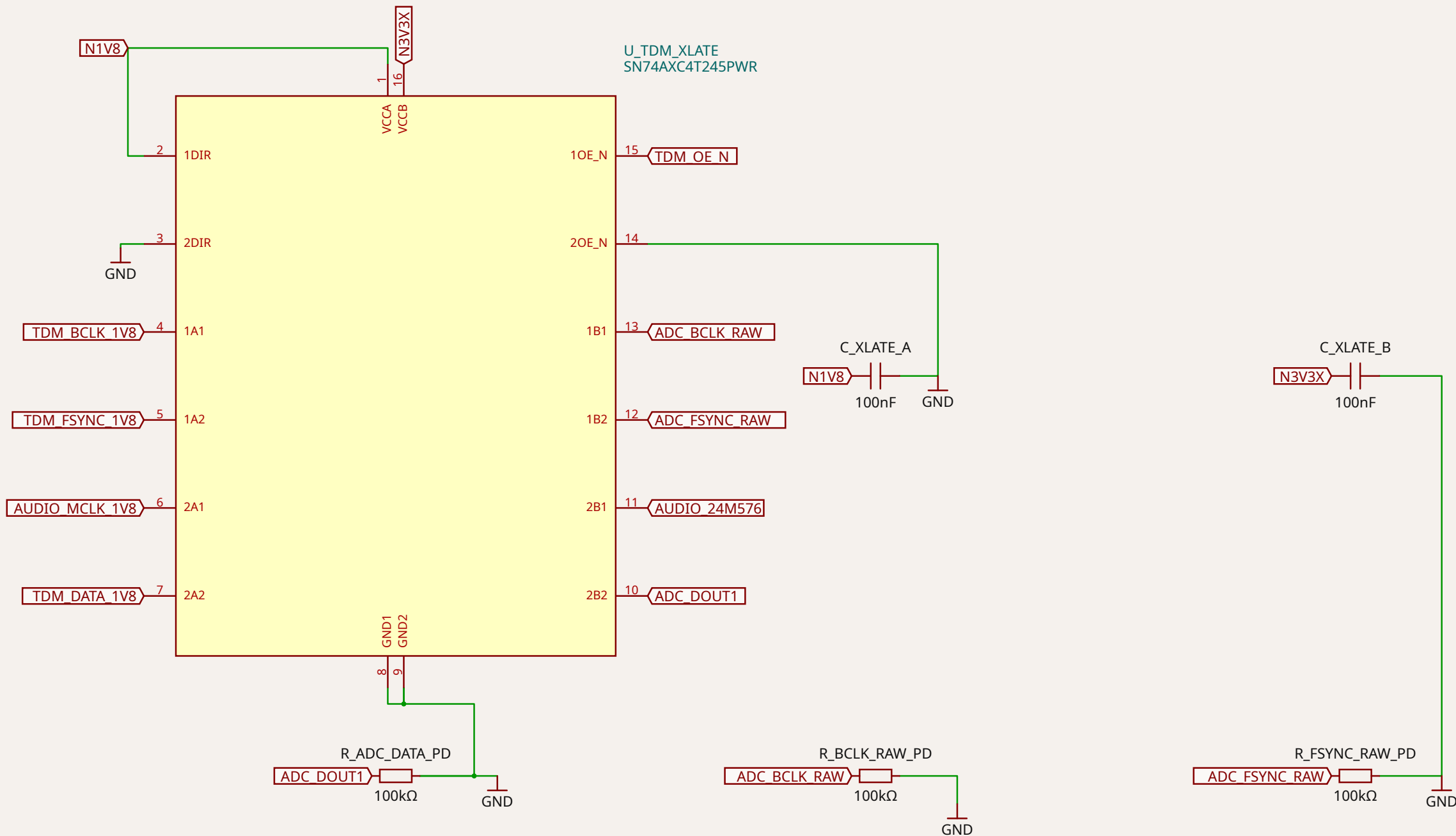
Page 33 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 2 components • landscape page fit



Crow USB Carrier v1

TDM TRANSLATION

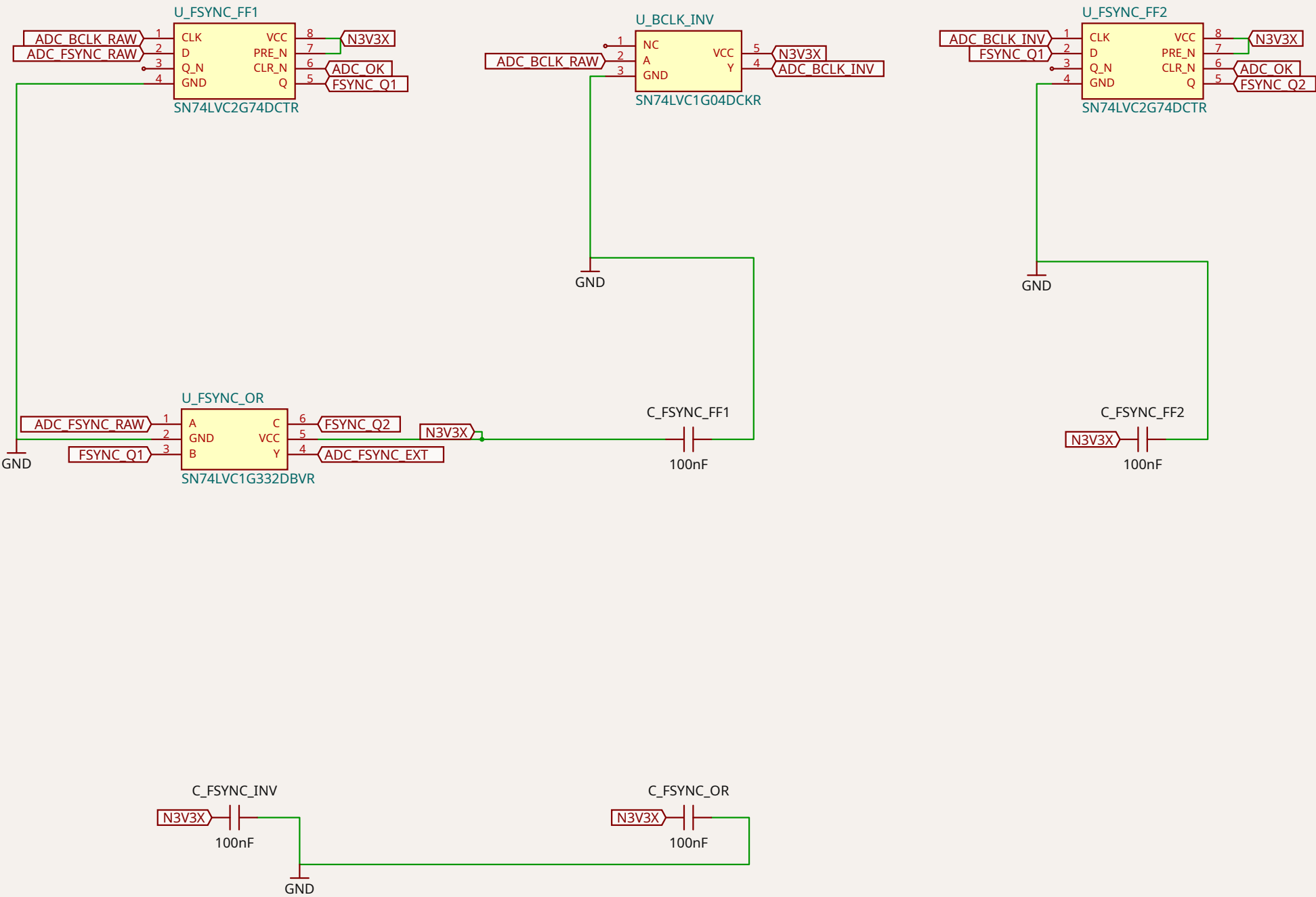
Page 34 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 6 components • landscape page fit



Crow USB Carrier v1

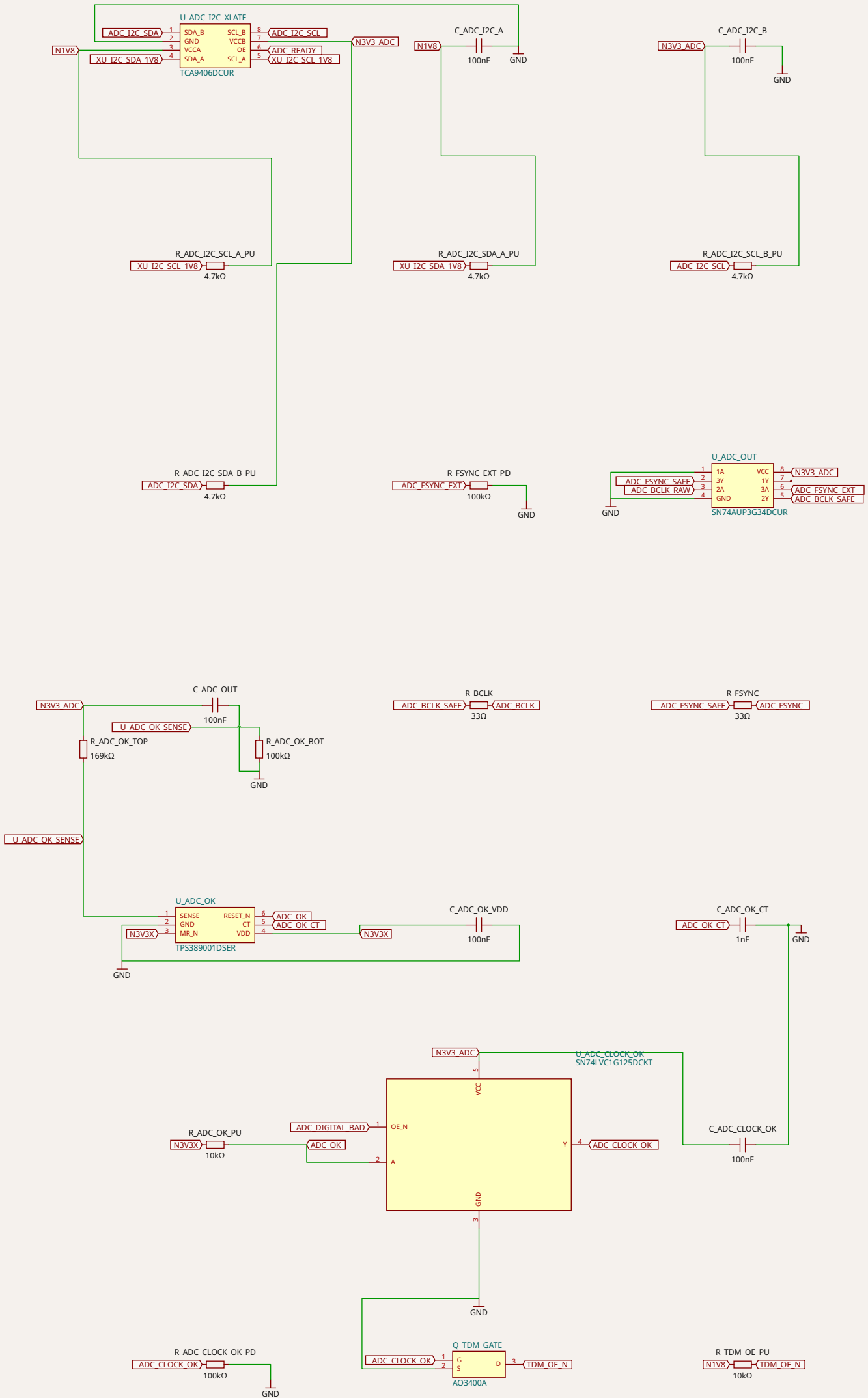
FSYNC SHAPING

Page 35 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 8 components • landscape page fit



Crow USB Carrier v1

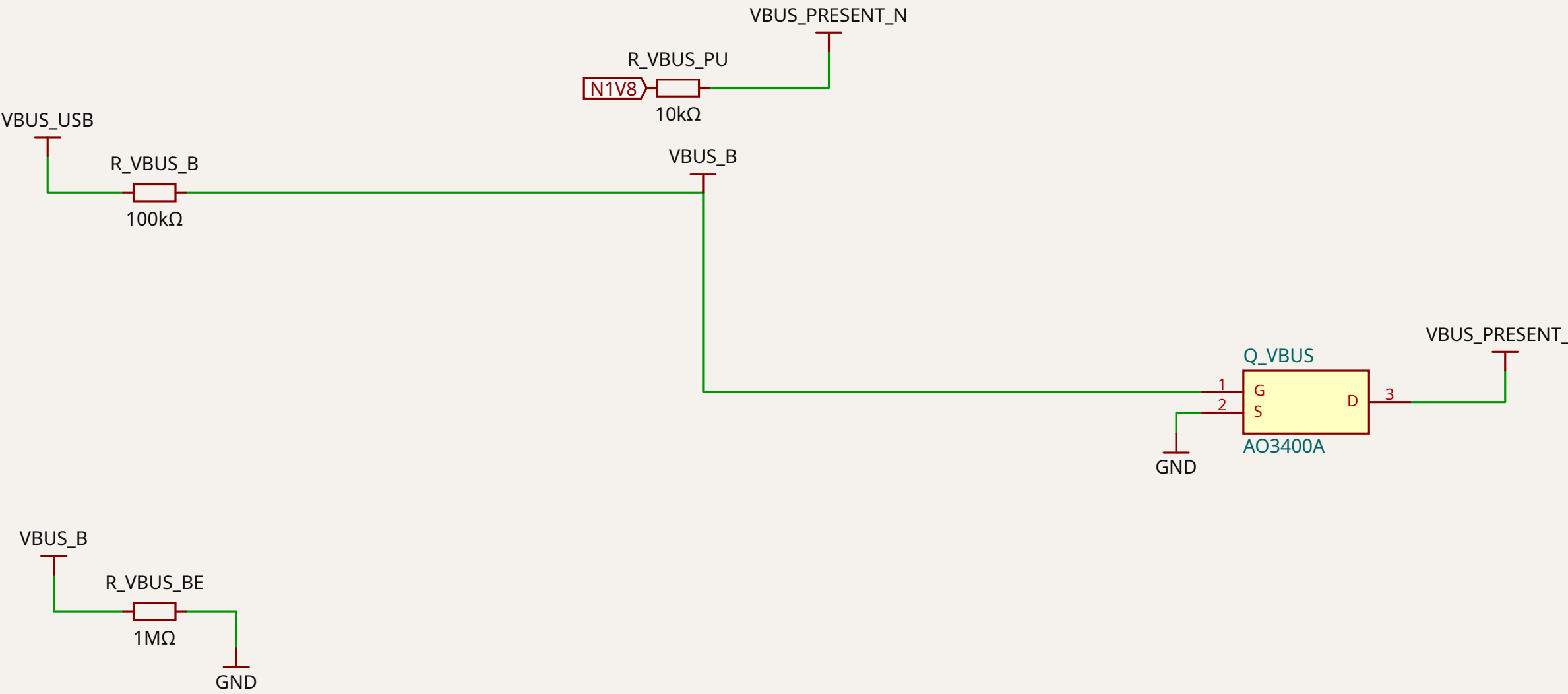
ADC CLOCK CONTROL



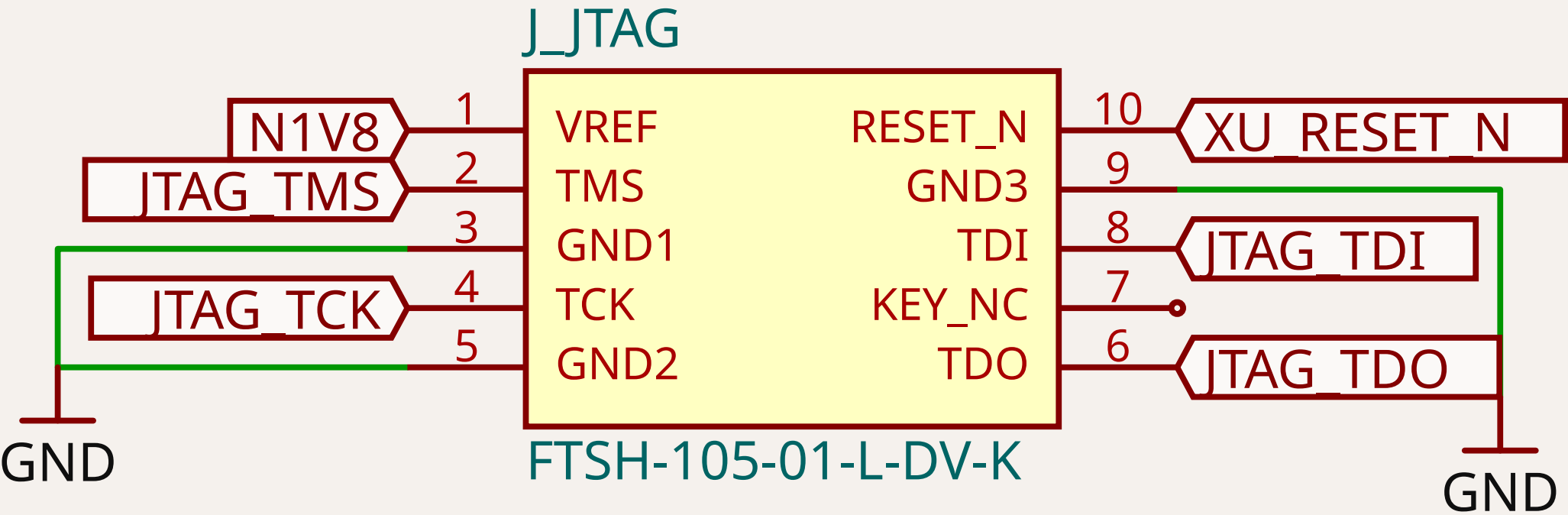
Crow USB Carrier v1

USB LOGIC

Page 37 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 4 components • landscape page fit



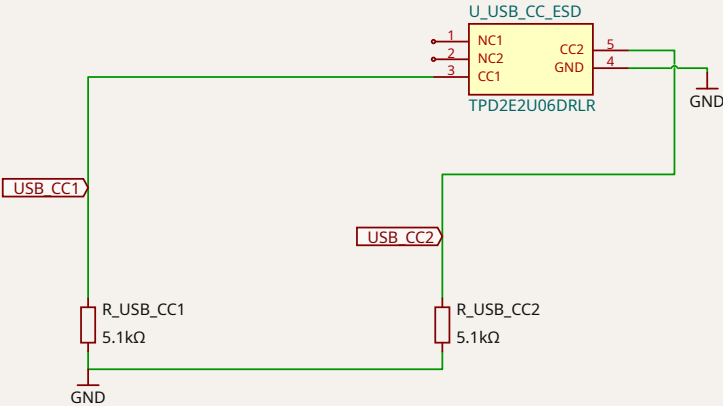
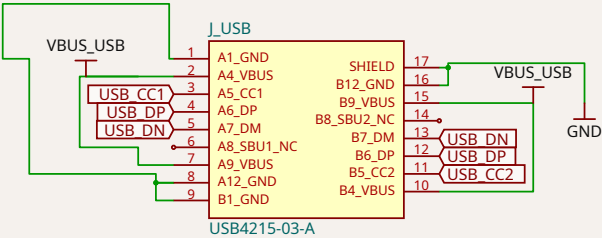
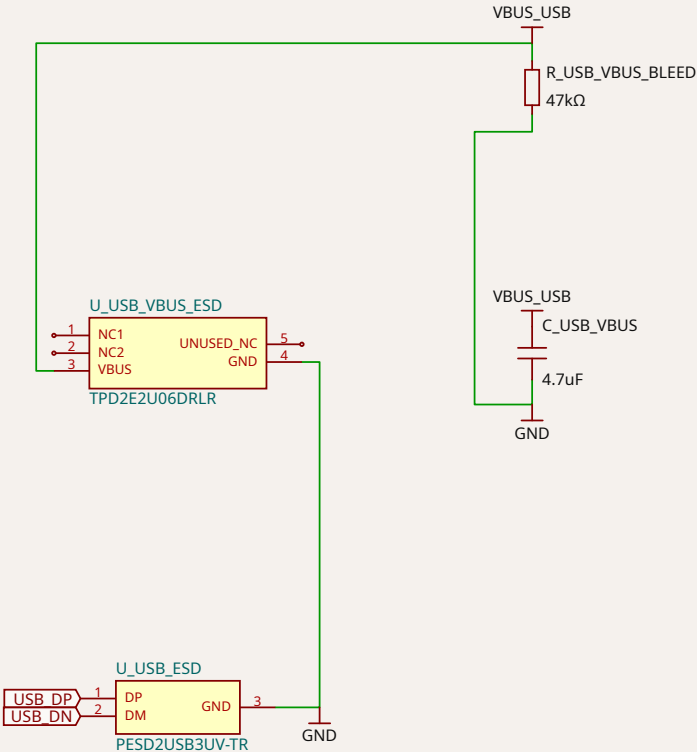
Crow USB Carrier v1
DEBUG



Crow USB Carrier v1

USB FRONTEND

Page 39 of 45 • exact circuit.json SHA-256 ff30fe8a46bc44be... • 8 components • portrait page fit



Semantic package-pin assignment appendix

Overview: XMO5 CORE — source sheet 30 / schematic_sheet_29

Circuit JSON SHA-256: ff30fe8a46bc44befa7533038bbf17d2ed0aed2e2b263b87dbd36f8fb5f3d41a

Native netlist SHA-256: 43132fc71a715d230738a65e9acbc89f55030e53a4541c34468127c048331c93

Pin tuples SHA-256: 0f84b3416445350b3c6cc18c9ea49cda78170d5e74a881ceb1d41244ea4b2e29

129/129 source pins verified against the fresh native netlist

- 4 audio / TDM
- 2 crystal
- 5 debug / reset
- 6 GND / ground
- 1 GPIO / VBUS sense
- 2 I2C
- 73 NC
- 26 power rail
- 6 QSPI flash
- 2 unused MIPI supply — grounded per XMO5 §14
- 2 USB

Crow USB Carrier v1

XMOS PIN INDEX — U_XU

U_XU package pins 1-26 of 129 — semantic assignment, not a coordinate crop

Pin	Package pin name	Net / NC	Function
1	X0D06	QSPI_D2	QSPI flash
2	X0D01	QSPI_CS_N	QSPI flash
3	X0D07	QSPI_D3	QSPI flash
4	X0D10	QSPI_CLK	QSPI flash
5	VDD	N0V9	power rail
6	X0D00_NC	NC	NC
7	X0D11_NC	NC	NC
8	X0D14	VBUS_PRESENT_N	GPIO / VBUS sense
9	X0D16_NC	NC	NC
10	VDDIOL	N1V8	power rail
11	VDD	N0V9	power rail
12	X1D36_NC	NC	NC
13	X1D37_NC	NC	NC
14	VDD	N0V9	power rail
15	X1D38_NC	NC	NC
16	X1D39_NC	NC	NC
17	VDDIOL	N1V8	power rail
18	VDD	N0V9	power rail
19	X1D00_NC	NC	NC
20	X1D01	TDM_FSYNC_1V8	audio / TDM
21	X1D09_NC	NC	NC
22	X1D10	TDM_BCLK_1V8	audio / TDM
23	X1D11	AUDIO_MCLK_1V8	audio / TDM
24	MIPI_VDD18	GND	unused MIPI supply — grounded per XMOS §14
25	MIPI_DN2_NC	NC	NC
26	MIPI_DP2_NC	NC	NC

Crow USB Carrier v1

XMOS PIN INDEX — U_XU

U_XU package pins 27–52 of 129 — semantic assignment, not a coordinate crop

Pin	Package pin name	Net / NC	Function
27	MIPI_VDD09	GND	unused MIPI supply — grounded per XMOS §14
28	MIPI_DN1_NC	NC	NC
29	MIPI_DP1_NC	NC	NC
30	VSS	GND	GND / ground
31	MIPI_DN0_NC	NC	NC
32	MIPI_DP0_NC	NC	NC
33	XOUT	XTAL_OUT	crystal
34	XIN	XTAL_IN	crystal
35	VDDI0B18	N1V8	power rail
36	TDI	JTAG_TDI	debug / reset
37	TD0	JTAG_TD0	debug / reset
38	RST_N	XU_RESET_N	debug / reset
39	VDD	N0V9	power rail
40	LV_L_N	GND	GND / ground
41	PLL_AVDD	PLL_0V9	power rail
42	PLL_AGND	GND	GND / ground
43	LV_T_N	GND	GND / ground
44	TMS	JTAG_TMS	debug / reset
45	VDD	N0V9	power rail
46	X0D12_NC	NC	NC
47	X0D13_NC	NC	NC
48	X0D22_NC	NC	NC
49	X0D23_NC	NC	NC
50	VDD	N0V9	power rail
51	TCK	JTAG_TCK	debug / reset
52	LV_R_N	GND	GND / ground

Crow USB Carrier v1

XMOS PIN INDEX — U_XU

Page 43 of 45 • overview page 30 • source sheet schematic_sheet_29 • circuit ff30fe8a46bc44be... • native 43132fc71a715d23...

U_XU package pins 53–78 of 129 — semantic assignment, not a coordinate crop

Pin	Package pin name	Net / NC	Function
53	X1D12_NC	NC	NC
54	VDD	N0V9	power rail
55	NC_NC	NC	NC
56	VDDI0B18	N1V8	power rail
57	X1D23_NC	NC	NC
58	USB_ID_NC	NC	NC
59	USB_DN	USB_DN	USB
60	USB_DP	USB_DP	USB
61	USB_VDD33	N3V3X	power rail
62	USB_VDD18	N1V8	power rail
63	X1D14_NC	NC	NC
64	X1D13_NC	NC	NC
65	X1D15_NC	NC	NC
66	X1D16_NC	NC	NC
67	X1D17_NC	NC	NC
68	VDD	N0V9	power rail
69	X1D18_NC	NC	NC
70	X1D19_NC	NC	NC
71	X1D20_NC	NC	NC
72	VDDI0R	N1V8	power rail
73	X1D21_NC	NC	NC
74	X1D22_NC	NC	NC
75	X1D49_NC	NC	NC
76	X1D50_NC	NC	NC
77	X1D51_NC	NC	NC
78	X1D52_NC	NC	NC

Crow USB Carrier v1

XMOS PIN INDEX — U_XU

U_XU package pins 79–104 of 129 — semantic assignment, not a coordinate crop

Pin	Package pin name	Net / NC	Function
79	X1D53_NC	NC	NC
80	X1D54_NC	NC	NC
81	X1D55_NC	NC	NC
82	X1D56_NC	NC	NC
83	X1D57_NC	NC	NC
84	X1D58_NC	NC	NC
85	VDD	N0V9	power rail
86	X0D24_NC	NC	NC
87	X0D25_NC	NC	NC
88	X0D26_NC	NC	NC
89	VDDI0R	N1V8	power rail
90	X0D27_NC	NC	NC
91	X0D28_NC	NC	NC
92	X0D29_NC	NC	NC
93	X0D35_I2C_SCL	XU_I2C_SCL_1V8	I2C
94	X0D36_I2C_SDA	XU_I2C_SDA_1V8	I2C
95	VDD	N0V9	power rail
96	X0D37_NC	NC	NC
97	X0D38_NC	NC	NC
98	X0D40_NC	NC	NC
99	X0D39_NC	NC	NC
100	X0D42_NC	NC	NC
101	X0D41_NC	NC	NC
102	X1D43_NC	NC	NC
103	X0D43_NC	NC	NC
104	VDD	N0V9	power rail

Crow USB Carrier v1

XMOS PIN INDEX — U_XU

U_XU package pins 105–129 of 129 — semantic assignment, not a coordinate crop

Pin	Package pin name	Net / NC	Function
105	VDD	N0V9	power rail
106	VDD	N0V9	power rail
107	X1D24	TDM_DATA_1V8	audio / TDM
108	X1D25_NC	NC	NC
109	VDDI0T	N1V8	power rail
110	X1D26_NC	NC	NC
111	X1D27_NC	NC	NC
112	X1D28_NC	NC	NC
113	VDD	N0V9	power rail
114	X1D29_NC	NC	NC
115	X1D30_NC	NC	NC
116	X1D31_NC	NC	NC
117	X1D32_NC	NC	NC
118	X1D33_NC	NC	NC
119	X1D34_NC	NC	NC
120	X1D35_NC	NC	NC
121	VDDI0T	N1V8	power rail
122	X0D30_NC	NC	NC
123	X0D32_NC	NC	NC
124	X0D31_NC	NC	NC
125	X0D34_NC	NC	NC
126	X0D33_NC	NC	NC
127	X0D04	QSPI_D0	QSPI flash
128	X0D05	QSPI_D1	QSPI flash
129	EP	GND	GND / ground